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Bow-Tie Architecture in the *Drosophila melanogaster* Connectome: A Two-Scale Structural Analysis

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Declaration

I hereby declare and confirm that this thesis is entirely the result of my own original work. Where other sources of information have been used, they have been indicated as such and properly acknowledged. I further declare that this or similar work has not been submitted for credit elsewhere.

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Abstract

Understanding how cognitive functions emerge from the physical wiring of the brain is one of the central challenges of modern neuroscience. The recent release of the synapse-resolution connectome of the adult *Drosophila melanogaster* brain has made it possible, for the first time, to address this question computationally on a complete nervous system.

This thesis asks whether the *Drosophila* connectome exhibits a bow-tie, or hourglass, architecture: whether many input pathways converge onto a small set of intermediate elements which then diverge again towards many outputs. The question is addressed at two complementary scales that serve as a check on each other. At the local scale we count, exhaustively and exactly, motifs in which k_{in} groups of neurons converge onto a single intermediate group, called the waist, which in turn projects onto k_{out} further groups. The counting uses a flow-vector formulation that never enumerates the search space, which is what makes an exhaustive treatment possible at all. At the global scale we compute the τ-core of the network, the smallest set of neurons carrying a prescribed fraction of all sensory-to-motor paths, and from it the hourglass score defined by Sabrin et al.

The thesis makes four contributions. The first is theoretical: we characterise exactly when motif counts compose, showing that the boundary between patterns countable with a linear flow vector and patterns requiring a sum-of-products formulation is precisely the presence of branching, and that cyclic patterns are not factorisable over vectors at all. The second is computational: a search engine that makes exhaustive counting tractable at connectome scale, gaining a factor of 219 on the worst bottleneck through a sparse reformulation, and verified row by row against an independent reference implementation. The third is empirical: an analysis of all four level windows of the hierarchy rather than a single one, which reveals that the architecture changes qualitatively with depth, with signatures involving a purely lateral branch growing from 20.1% to 76.2% of the motif mass. The fourth is methodological: a negative result, in which an apparent neurochemical difference between feedback and feed-forward waists proves not to survive replication across independent windows and is traced to regional composition.

The hourglass effect is present but depends on how paths are routed, with a score of 0.498 under structural routing against values between 0.74 and 0.87 reported for *C. elegans*. The two scales agree on the mid-hierarchy waists of the olfactory pathway and disagree on input groups, and that disagreement turns out to be interpretable rather than contradictory.

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Glossary

AL	Antennal Lobe, the first olfactory processing centre
BLAS	Basic Linear Algebra Subprograms
E/I	Excitatory / Inhibitory
EB	Ellipsoid Body
EM	Electron Microscopy
FB	Feedback, an edge running from a deeper to a shallower level
FDR	False Discovery Rate
FF	Feed-forward, an edge running from a shallower to a deeper level
GNG	Gnathal Ganglia
lat	Lateral, an edge between neurons of the same level
LH	Lateral Horn
LO	Lobula, a visual processing neuropil
LOP	Lobula Plate
MB	Mushroom Body, the centre for olfactory learning
MB_CA	Mushroom Body Calyx, its input region
ME	Medulla, the largest visual neuropil
PLP	Posterior Lateral Protocerebrum
PRW	Prow
PVLP	Posterior Ventrolateral Protocerebrum
$\Sigma\Pi$	The sum-of-products formulation used for branching patterns

1 Introduction

1.1 Motivation

One of the most ambitious goals of modern neuroscience is to understand how cognitive functions such as memory, learning and decision-making emerge from the physical structure of the brain [1]. For most of the history of the field this question could only be approached indirectly. Lesion studies, electrophysiology and imaging observed neural activity without access to the underlying wiring diagram, which meant that structural hypotheses about brain function could be formulated but not rigorously tested.

Advances in large-scale electron microscopy and automated image segmentation have begun to dissolve this barrier. Connectomics, the science of producing complete maps of synaptic connectivity in neural tissue [3], now yields global wiring diagrams in which every neuron and every synapse is labelled. Where traditional neural sampling reconstructed a few hundred neurons at a time, a connectome provides the entire network at once, which permits computational analyses at a scale and resolution that were previously out of reach. The adult *Drosophila melanogaster* brain has emerged as the premier model organism for this pursuit: it contains only about 130,000 neurons, some five orders of magnitude fewer than the human brain, and yet supports a behavioural repertoire spanning visual pursuit, olfactory learning, sleep regulation and courtship [7].

The availability of a complete directed graph of a complex brain creates an opportunity, but it does not by itself create understanding. A raw connectome is a sparse, cyclic and heterogeneous directed graph with no intrinsic coordinate system. Artificial neural networks are designed with layered topologies through which information flows in one direction; biological circuits are not. They feature extensive recurrent connectivity, and feedback projections from deeper processing stages back to earlier ones are plentiful relative to feed-forward ones [13]. This makes it structurally difficult to say what “earlier” and “later” mean in a circuit, and consequently difficult to compare connectivity patterns across brain regions in any principled way, because there is no agreed axis along which to place them.

A central concept for making sense of this complexity is that of network motifs: small recurring connectivity patterns that occur in a network far more often than chance would predict [9]. At the level of neural circuits, motifs are the elementary computational building blocks from which larger structures are assembled. A feed-forward chain of three neurons implements a simple relay; adding a feedback arc to that chain turns it into a loop capable of sustaining activity over time; a lateral connection between two populations firing in parallel creates a cross-stream interaction. If one knows which motifs are common in a given connectome, and how their composition changes from one brain region to another, it becomes possible to read circuit-level computational strategies off the wiring diagram alone.

This thesis focuses on one particular architectural template, the bow-tie, also called the hourglass. A bow-tie is a structure in which many inputs converge onto a narrow set of intermediate elements, called the waist, which then diverge again towards many outputs. The template is known to recur across widely different complex systems: cellular metabolism, the Internet protocol stack and manufacturing supply networks all display it [10]. Its presence is informative because it implies that the system compresses information through a narrow channel before redistributing it, and that in turn constrains both what the system can compute and how vulnerable it is to the loss of individual components. If the *Drosophila* brain is organised this way, the neurons occupying the waist are the ones through which most of the sensory-to-motor traffic must pass, which makes them natural targets for functional investigation.

1.2 Problem Statement

The technical question of this thesis can be stated as follows. Given the full synaptic graph of the adult *Drosophila* brain, comprising 134,181 neurons with an assigned hierarchical level and 2,700,513 synaptic pairs, determine whether the network exhibits a bow-tie architecture, at what scale it does so, and which anatomical structures occupy the waist. The problem decomposes into three sub-problems, each of which has to be solved before the next becomes meaningful.

The first is that of counting without enumeration. A bow-tie motif with two input groups and two output groups, searched over the groups of a single level window, induces a space of more than eight billion candidate combinations. Explicit enumeration of subgraph occurrences, in the style of the classical subgraph isomorphism algorithms [12, 4], is entirely infeasible at that scale. What is required instead is a counting formulation that produces exact occurrence counts without ever materialising an individual match. Such formulations do exist, but they do not exist for every pattern, and the conditions under which they apply have to be established rather than assumed. Chapter 3 shows that the condition is a property of the shape of the pattern and nothing else.

The second sub-problem concerns the relationship between two different ways of asking the same question. Motif counting is inherently local: it measures convergence and divergence in the immediate neighbourhood of a group of neurons. The hourglass effect, by contrast, is a global property of end-to-end information flow through the whole network. These are two different measurements of the same architectural intuition, and there is no guarantee that they agree. Establishing whether they do, and characterising precisely where they do not, is part of the problem rather than an afterthought, because a disagreement between two independent measurements is either a warning that one of them is wrong or a finding in its own right.

The third sub-problem is that of distinguishing structure from composition. Anatomical regions of the fly brain differ systematically in size, in connection density and in neurotransmitter composition. Any apparent difference between two classes of motifs may therefore reflect merely *where* those motifs happen to live rather than *what* they are. Controlling for regional composition is essential, and as Section 4.7 demonstrates in detail, failing to replicate such a control across independent subsets of the data can produce results that look convincingly significant and are not.

1.3 Research Objectives

The first objective is to establish the algebra of pattern counting, by determining which connectivity patterns admit a linear counting formulation, which require a sum-of-products formulation, and which require quadratic memory. The result should be a statement about the shape of the pattern rather than an empirical rule of thumb, and it should be verifiable by exhaustive enumeration on small graphs so that its correctness does not rest on the derivation alone.

The second objective is to make exhaustive counting tractable, by designing and implementing a search engine capable of counting all bow-tie motifs over the full connectome, for every level window, in hours rather than days. Here correctness must be verified against an independent reference implementation, not merely inferred from the mathematics, because an optimisation that is correct in principle can still be wrong in code.

The third objective is to characterise the architecture at both scales, by applying the local motif search and the global hourglass analysis to the same connectome, reporting where the two converge and where they diverge, and interpreting the divergence rather than explaining it away.

The fourth objective is to test structural hypotheses with adequate controls. This means evaluating the statistical significance of the observed motifs against several complementary null models rather than one, and testing biological hypotheses about the waists, in particular whether feedback and feed-forward waists differ in neurotransmitter composition, with explicit controls for regional composition and with replication across independent level windows.

1.4 Contributions

The theoretical contribution of this thesis is an exact characterisation of when motif counts compose, developed in Section 3.1.3. Counting occurrences of a pattern means counting homomorphisms from that pattern into the graph, and both the cost of doing so and the way counts combine depend only on the shape of the pattern. Paths admit a pure matrix product and their counts add across unions of node groups; trees require an element-wise product at every branching node and their counts do not add; cyclic patterns cannot be factorised over vectors at all. The boundary between the tractable and the intractable is therefore exactly the presence of branching, and an exhaustive catalogue up to four nodes shows that 87% of connected patterns already fall on the intractable side.

The computational contribution is a search engine that is both fast and verified, described in Section 3.5. Reformulating the inner loop as a single matrix product delegated to optimised linear algebra routines, and then switching to a sparse product for the largest waists, yields a factor of 219 on the worst bottleneck of the search. The engine is verified row by row, across all 1,043,872 rows and all 26 columns, against a slow but readable reference implementation retained specifically for that purpose.

The empirical contribution is an analysis across all four level windows of the hierarchy, presented in Section 4.3. Previous work on this connectome has concentrated on a single window. Extending the analysis to all four shows that the structure changes qualitatively with depth: signatures involving a purely lateral branch carry 20.1% of the motif mass in the shallow window and 76.2% in the deep one, and the deepest level of the hierarchy turns out to be the most internally dense while remaining marginal as a waist.

The methodological contribution is a negative result, established properly rather than quietly dropped, and reported in Section 4.7. The hypothesis that feedback waists carry a distinct neurochemical signature is significant on one window, reverses sign on another, and vanishes on the full connectome. The apparent effect is traced to regional segregation, and the procedure that exposed it is described in enough detail to be reused.

1.5 Thesis Outline

Section 2 introduces the background required for the rest of the work: the graph formalism, the *Drosophila* connectome and its hierarchical decomposition, network motifs, and the hourglass framework of Sabrin et al. Section 3 develops the counting theory, both analysis pipelines, the statistical validation scheme and the computational optimisations that make the analysis feasible. Section 4 presents the results, and Section 5 summarises the contributions, states the limitations and outlines directions for further work.

2 Theoretical Background

2.1 Fundamentals of Network Science

2.1.1 Graph Formalism

A connectome is modelled as a directed graph $G = (V, E)$, where V is the set of neurons and $E \subseteq V \times V$ the set of synaptic connections. An edge $(u, v) \in E$ indicates that neuron u forms at least one chemical synapse onto neuron v; the direction of the edge is the direction in which the signal travels. Edges may additionally carry a weight $w(u, v)$ equal to the number of individual synaptic contacts between the two neurons, which in this dataset ranges from the reporting threshold of five up to several hundred.

Throughout this thesis we work with the unweighted, de-duplicated graph, so that an ordered pair of neurons contributes at most one edge regardless of how many synapses connect them. This choice is deliberate and worth justifying, because it discards information. Motif counting asks whether a connectivity *pattern* exists between particular neurons, not how strong that connection is. If multiplicities were retained, a motif count would conflate the number of distinct circuits realising a pattern with the strength of the individual connections within them, and a single very strongly connected pair could dominate the count of a pattern that occurs only once. Since the quantity of interest here is how many distinct circuits implement a given architecture, the unweighted graph is the correct object.

2.1.2 Matrix Representations

The adjacency matrix $A \in \{0, 1\}^{|V| \times |V|}$ has $A_{uv} = 1$ if and only if $(u, v) \in E$. For a connectome of 134,181 neurons a dense representation of this matrix would occupy more than 18 GB at one byte per entry, whereas the graph contains only 2,700,513 edges, a density below 1.5×10^{-4}. Sparse representations, and in particular the Compressed Sparse Row format, are therefore not an optimisation but a prerequisite for working with the data at all.

The reason the matrix representation matters for this thesis is a single identity. The number of directed paths of length k running from a set of neurons S to a set of neurons T is given by

$$\mathbf{1}_S^\top A^k \, \mathbf{1}_T, \tag{1}$$

where $\mathbf{1}_S$ is the indicator vector of the set S. What makes this useful is that the expression need never be evaluated by forming A^k, which would be dense and enormous. Instead one propagates a vector through k successive sparse matrix–vector products, at a cost of $O(k \cdot |E|)$ time and $O(|V|)$ memory. An entire class of counting problems thus becomes linear in the size of the network. Section 3.1.3 establishes precisely how far this identity extends, and where it stops.

2.2 The *Drosophila melanogaster* Connectome

2.2.1 A Model Organism for Structural Analysis

The adult *Drosophila* brain occupies a favourable position in the space of model organisms. It is small enough to have been reconstructed completely at synaptic resolution, which no vertebrate brain has been, and yet complex enough to support behaviour that is not trivially reflexive. With approximately 130,000 neurons it retains sensory integration, associative learning and memory, so that structural findings about it are plausibly informative about principles that generalise, rather than being peculiarities of a simple nervous system.

2.2.2 The FlyWire Dataset

The data used in this work derive from the FlyWire reconstruction of the adult female brain [7], distributed through the Codex portal. Four tables are used. The first lists synaptic pairs with their contact counts, already filtered at a threshold of five synapses in the public export, which removes the least reliable segmentation artefacts. The second gives neuron identifiers together with their anatomical group assignment and their predicted neurotransmitter. The third provides functional super-class labels, distinguishing sensory, optic, central, descending and visual projection neurons. The fourth supplies the hierarchical level assignment described below.

After restricting the data to neurons that possess both a level assignment and a non-degenerate anatomical group, the working network comprises 134,181 neurons and 2,700,513 de-duplicated synaptic pairs. All figures reported in this thesis refer to that restricted network.

2.2.3 Hierarchical Levels

The raw connectome has no intrinsic notion of depth, and without one it is impossible to say whether a given connection runs forwards or backwards. To provide such a notion, each neuron is assigned a discrete hierarchical level $y \in \{1, \ldots, 5\}$ reflecting its position along the axis running from sensory input to motor output. Level 1 contains the neurons closest to the sensory periphery and level 5 those closest to motor output. The level assignment used here is taken as given from earlier work on this dataset; this thesis does not re-derive it, but it does validate it independently in Section 4.1, where the functional composition of the levels is shown to follow the expected gradient.

Once levels are available, every edge acquires a direction relative to the hierarchy. For an edge running from a neuron at level y_u to a neuron at level y_v we write

$$\mathrm{dir}(u, v) = \begin{cases} \text{FF} & \text{if } y_u < y_v \quad \text{(feed-forward)}, \\ \text{FB} & \text{if } y_u > y_v \quad \text{(feedback)}, \\ \text{lat} & \text{if } y_u = y_v \quad \text{(lateral)}. \end{cases} \tag{2}$$

A feed-forward edge carries information deeper into the hierarchy, a feedback edge returns it towards the periphery, and a lateral edge moves it sideways within a single processing stage. This three-way classification is used throughout the thesis and underlies the finer signature scheme introduced in Section 2.3.

2.2.4 Group-Level Abstraction

Counting motifs over individual neurons is both computationally hopeless and biologically uninformative. It is hopeless because a five-node motif over 134,000 nodes has astronomically many candidate instantiations, and uninformative because individual neurons are not the units at which circuit function is usually described: what one wants to know is that the antennal lobe projects to the mushroom body calyx, not that one particular neuron does.

We therefore abstract from neurons to *metanodes*. A metanode is a pair consisting of an anatomical group and a hierarchical level, so that for example the neurons of the mushroom body calyx sitting at level 2 form one metanode, and those of the same structure at level 3 form another. A motif is then a pattern over metanodes, and its count is the number of distinct neuron-level instantiations of that pattern. This preserves the quantity that actually matters, namely how many real circuits realise the pattern, while reducing the search space by several orders of magnitude. Depending on the level window under consideration, the abstraction yields between

879 and 1,220 metanodes, a number small enough that an exhaustive search over combinations of them becomes conceivable.

2.3 Network Motifs

2.3.1 Motifs as Building Blocks

Milo et al. [9] introduced network motifs as patterns of interconnection that recur in a network significantly more often than they do in randomised networks preserving specified properties. The significance requirement is what distinguishes a motif from a merely frequent subgraph, and it is essential: a pattern may be common simply because the network is dense, or because certain nodes happen to have very high degree, without that frequency reflecting anything about the organisation of the system. The comparison against a randomised ensemble is therefore not a statistical formality added at the end but part of the definition, and Section 3.4 describes the three ensembles used here.

2.3.2 Directional Classification

Given the level assignment, every edge of a motif carries one of the three directions of Equation (2). A bow-tie motif with two input groups and two output groups has four edges, two entering the waist and two leaving it. Rather than labelling the motif by all four edges individually, which would produce an unwieldy number of categories, we classify each *branch* by whether its edges agree with one another. If both edges entering the waist are feed-forward, the input branch is labelled FF; if they disagree, it is labelled *mix*. The same applies to the output branch. The resulting label is written

$$\text{signature} = \text{sig}_{\text{in}} \to \text{sig}_{\text{out}}, \qquad \text{sig} \in \{\text{FF}, \text{FB}, \text{lat}, \text{mix}\}, \tag{3}$$

which yields sixteen possible signatures. A motif labelled `FF->lat`, for instance, is one in which two groups feed forward onto a waist that then distributes laterally within its own level.

A coarser classification was used in earlier work on this dataset. It labels a motif *pure_FF*, *pure_FB* or *pure_lateral* when all four of its edges agree, and *mixed* otherwise. As Section 4.3.2 shows quantitatively, this coarse label turns out to be close to vacuous: with four edges, complete agreement is improbable, and 94.4% of all patterns consequently fall into the *mixed* bucket. The sixteen-signature scheme resolves that bucket into classes that behave very differently from one another, and much of the analysis in this thesis depends on the finer resolution.

2.4 Bow-Tie and Hourglass Architecture

2.4.1 The Bow-Tie Motif

Definition 2.1 (Bow-tie motif)**.** Given metanodes $A_1, \ldots, A_{k_{\text{in}}}$, a waist metanode B, and metanodes $D_1, \ldots, D_{k_{\text{out}}}$, with $\{A_m\} \cap \{D_n\} = \emptyset$ and $B \notin \{A_m\} \cup \{D_n\}$, a bow-tie motif instance is a tuple of distinct neurons $(a_1, \ldots, a_{k_{\text{in}}}, b, d_1, \ldots, d_{k_{\text{out}}})$ with $a_m \in A_m$, $b \in B$, $d_n \in D_n$, such that $(a_m, b) \in E$ for all m and $(b, d_n) \in E$ for all n.

Figure 1 shows the structure schematically. The requirement that the input and output groups be disjoint deserves particular emphasis, because it is easy to omit and expensive to omit. Without it a metanode may appear simultaneously among the inputs and the outputs of the same motif, so that what is counted is not a bow-tie at all but a configuration in which a group feeds back into itself through the waist. Such configurations are extremely numerous, because the same group reappearing on both sides multiplies the available combinations, and they are also degenerate in the sense that they do not describe convergence followed by divergence. As reported in Section 4.3, enforcing the constraint removes 6.60% of the patterns but 57.3% of the total count mass, which gives a measure of how strongly degenerate configurations dominate when the search is left unconstrained.

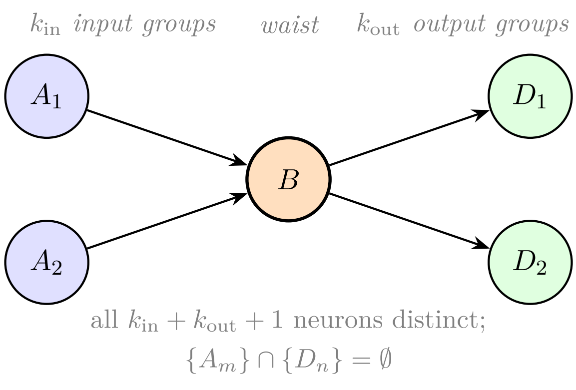

Figure 1: The bow-tie motif of Definition 2.1. Each node is a metanode, that is, a pair consisting of an anatomical group and a hierarchical level; an instance of the motif is a tuple of distinct neurons, one drawn from each metanode, connected as shown. The disjointness requirement between input and output groups is what prevents degenerate configurations in which a group feeds back into itself through the waist.

2.4.2 The Hourglass Effect

Sabrin et al. [10] formalise the hourglass property of a hierarchical network in three steps. One begins by fixing a set S of source nodes and a set T of target nodes; in a connectome these are naturally the sensory and the motor neurons respectively. For each node v of the network one then computes its *path centrality*, defined as the number of source-to-target paths that traverse v. Ordering all nodes by decreasing path centrality and taking the smallest prefix of that ordering which together covers a fraction τ of all source-to-target paths yields the τ*-core*, written $C(\tau)$. Intuitively, the τ-core is the smallest set of neurons through which most of the traffic must pass.

A small core is not by itself evidence of an hourglass, and this is the point at which the framework becomes subtle. A network might have a small core simply because its source-to-target dependencies are themselves concentrated, quite apart from any hierarchical organisation. The comparison is therefore made against the *flat dependency network*, constructed by connecting every source directly to every target it can reach and discarding all intermediate structure. This flat network has exactly the same dependencies as the original but no hierarchy whatsoever. Writing $C_f(\tau)$ for its core, the hourglass score is

$$H = 1 - \frac{|C(\tau)|}{|C_f(\tau)|}. \tag{4}$$

A value of H close to 1 means that the real network routes its flow through far fewer intermediaries than its dependency structure requires, which is the signature of an hourglass. A value close to 0 means that the apparent concentration is entirely explained by the dependencies themselves and that the hierarchy contributes nothing.

2.5 Related Work

Two recent theses prepared under the same supervision analyse the same connectome with complementary aims. Tricoli [11] studies neuronal assemblies through motif-based characterisation combined with hierarchical flow ranking, with the goal of identifying groups of neurons that share structural roles and might therefore act as functional units. Curcio [6] introduces a parametric framework for motif discovery conditioned on hierarchical position, and explores the resulting search space along three axes: the vertical depth of the window, the horizontal complexity of the pattern, and the distance spanned by skip connections.

Both of those works enumerate motifs of fixed size and interpret them in terms of super-class composition. The present thesis differs from them in three respects. It targets one specific architectural template, the bow-tie, rather than attempting a full motif census, and this restriction is what makes exact counting through a closed formulation possible where enumeration would not be. It pairs the local analysis with a global hourglass measurement and then quantifies the agreement between the two rather than reporting either in isolation. And it treats statistical validation and negative results as first-class outputs, on the view that an analysis whose controls are not reported cannot be assessed by a reader.

Earlier work on this connectome [5] treated the brain as a flat homogeneous graph and enumerated motifs without conditioning on hierarchical position at all, which produces useful global statistics but collapses together circuits operating at very different processing stages.

3 Methods

3.1 Counting Without Enumeration

3.1.1 Homomorphism Counting and Flow Vectors

Counting the occurrences of a pattern P in a graph G amounts to counting the homomorphisms from P into G that respect the prescribed group memberships. A homomorphism here is simply an assignment of a neuron to each node of the pattern such that every edge of the pattern is realised by an edge of the graph. For a pattern that is a *path* this reduces to Equation (1): one propagates a vector along the pattern through a sequence of sparse matrix–vector products, and the total count falls out at the end. The vector being propagated is called the *flow vector*, and its i-th entry has a direct interpretation as the number of partial matches of the pattern that terminate at neuron i.

The flow vector has a property that turns out to be decisive. Because the operation is linear, the count computed over a union of groups equals the sum of the counts computed over the individual groups. Counting the pattern on $A_1 \cup A_2$ therefore requires no more work than counting it on A_1 and on A_2 separately and adding the results. This is what allows a search to consider large collections of groups without the cost growing with the number of combinations of them.

3.1.2 The $\Sigma\Pi$ Formulation

A bow-tie is not a path. It is a tree with a branching point at the waist, and at that branching point the two incoming branches are required to be incident on the *same* neuron. That requirement is what destroys the linearity, because the count of the combined pattern is no longer determined by the counts of its parts: knowing how many neurons of A_1 reach the waist and how many neurons of A_2 reach it does not tell us how many neurons of the waist are reached by both.

The correct formulation must therefore work neuron by neuron across the waist. Let $v_{A_m}[i]$ denote the number of neurons of group A_m that project onto neuron i of the waist, and let $w_{D_n}[i]$ denote the number of neurons of group D_n that receive from that same neuron i. Then the exact count of the bow-tie is

$$c(A_1, \ldots, A_{k_{\text{in}}}; B; D_1, \ldots, D_{k_{\text{out}}}) \;=\; \sum_{i \in B} \left(\prod_{m=1}^{k_{\text{in}}} v_{A_m}[i] \right) \left(\prod_{n=1}^{k_{\text{out}}} w_{D_n}[i] \right). \tag{5}$$

Reading the expression from the inside outwards makes its logic clear. For a fixed waist neuron i, the product over m counts the ways of choosing one input neuron from each input group such that all of them project onto i, and the product over n counts the ways of choosing one output neuron from each output group such that i projects onto all of them. Multiplying the two gives the number of complete bow-ties centred on i. Summing over the waist then aggregates across all possible centres. The products enforce coincidence on a single neuron; the sum aggregates over the waist. It is from this structure that the formulation takes its name, $\Sigma\Pi$, and it is exact: no approximation and no sampling are involved anywhere.

3.1.3 The Algebra of Patterns

The contrast between Equations (1) and (5) is not an accident of the bow-tie. It generalises to a statement about pattern shapes.

Proposition 3.1 (Counting cost by pattern shape). *Let P be a connectivity pattern to be counted in a graph on n nodes. If P is a* path*, its count is a pure matrix product, computable in $O(n)$ memory, and the count is additive over unions of node groups. If P is a* tree*, its count requires matrix–vector products along the edges together with an element-wise product at each branching node; memory remains $O(n)$, but additivity fails at the branching nodes. If P is* cyclic*, its count is not factorisable over vectors at all, and a matrix must be maintained, requiring $O(n^2)$ memory.*

The practical consequence is that the boundary between the cheap and the expensive is exactly the presence of branching, and that a second boundary, between the feasible and the infeasible at connectome scale, is the presence of a cycle. This is not an empirical observation but a consequence of the structure of the pattern, and it was verified rather than assumed: for all connected patterns up to four nodes, the closed-form counts derived automatically from the pattern were compared against explicit enumeration on random graphs and agreed exactly in every case.

Proposition 3.1 also answers a question that arises naturally when one tries to combine circuits. Given the counts of two linear chains that share a middle node, when is the count of their combination obtained by composing them? The answer is that composition works along the chain and fails at the junction, and the junction is precisely where $\Sigma\Pi$ becomes necessary.

3.1.4 How Much Ground the Method Covers

Proposition 3.1 classifies patterns, but it does not by itself say how much of the space of patterns each class occupies, and without that the classification could be a distinction without practical consequence. To settle the question we enumerated all connected non-isomorphic patterns up to four nodes, derived the factorised counting formula automatically for each of them, and verified every formula against exhaustive homomorphism enumeration on random graphs.

Table 1: Exhaustive catalogue of connected patterns by shape, with formulae verified against explicit enumeration.

Nodes	Patterns	Paths	Trees	Cyclic	Formulae verified
3	13	6	0	7 (54%)	6/6
4	199	15	10	174 (87%)	25/25

All thirty-one derived formulae reproduced the enumerated counts exactly, which establishes that the automatic derivation is correct on every case it was tested against. The figure worth carrying forward, however, is in the penultimate column of the second row: already at four nodes, 87% of connected patterns are cyclic and therefore not factorisable over vectors. The flow-vector and $\Sigma\Pi$ machinery is thus powerful on the patterns it covers and silent on the large majority of the space. That is exactly why a bow-tie, which is a tree, is a tractable target at connectome scale while a general motif census is not, and it places the present method in relation to enumeration-based approaches rather than in competition with them.

Two boundaries of this result should be stated plainly. The catalogue stops at four nodes because exhaustive enumeration on five would require 2^{20} edge masks for 120 permutations, which this approach cannot reach; the classification itself does not depend on pattern size, so the qualitative conclusion carries, but the percentages beyond four nodes are not measured here. Furthermore, the classification concerns homomorphisms, whereas the requirement that the participating neurons all be distinct is a separate layer of inclusion–exclusion sitting on top of the count. The two should not be conflated: factorisability is a property of the shape of the pattern, whereas node distinctness is a property of the kind of count one wishes to perform.

3.2 Micro-Scale Analysis: Bow-Tie Motif Search

3.2.1 Search Space and Parameters

The search is governed by four parameters. The level window NL fixes the set of hierarchical levels within which motifs are sought, so that $NL = \{1, 2, 3\}$ restricts attention to early sensory processing while $NL = \{3, 4, 5\}$ examines deep integration. The parameters k_{in} and k_{out} set the number of input and output groups of the bow-tie. The parameter *max_jump* bounds the level difference that a single edge of the metagraph may span, and setting it to one imposes a layer-by-layer processing logic in which a connection may move at most one level up or down. Finally *min_count* sets the minimum number of occurrences a pattern must have to be retained, which suppresses the long tail of patterns realised by only a handful of neurons. Unless stated otherwise, all results in this thesis use $k_{\text{in}} = k_{\text{out}} = 2$, $max\_jump = 1$ and $min\_count = 10$.

3.2.2 Pruning

Three constraints reduce the search space, and it is important that none of them affects exactness: each removes only combinations that provably contribute nothing. The first is the disjointness requirement of Definition 2.1, which excludes combinations in which an input group also appears among the outputs. The second is a restriction to the support of the waist: only those waist neurons that are reachable from at least one input group and that in turn reach at least one output group can possibly contribute to a count, so all others are removed before any product is formed. The third is an early exit on null flow, which discards any combination in which one of the factors of Equation (5) is identically zero, since the product is then zero regardless of the remaining factors. On the shallow window these three together eliminate more than 97% of the nominal search space.

3.2.3 Compression Metrics

A waist is interesting to the extent that it *compresses*, that is, to the extent that many peripheral neurons are routed through few central ones. The obvious way to measure this is to take the ratio of the number of peripheral neurons to the size of the waist group, but that measure fails in practice, and understanding why it fails motivates the correct one.

The difficulty is that a group may be nominally large while only a handful of its neurons actually participate in any given pattern. A waist of twenty neurons of which two carry all the flow behaves, functionally, like a waist of two, and dividing by twenty understates its compression by an order of magnitude; conversely, a group of twenty neurons that all participate equally is a genuinely distributed bottleneck. What is needed is a measure of the *effective* number of participating neurons, one that equals the group size when all neurons contribute equally and falls towards one when a single neuron dominates.

The participation ratio provides exactly that. Writing P_i for the $\Sigma\Pi$ contribution of waist neuron i, that is, the number of bow-ties centred on that neuron, we define

$$n_{\text{waist}}^{\text{eff}} = \frac{\left(\sum_i P_i\right)^2}{\sum_i P_i^2}, \qquad \text{realised compression} = \frac{\text{peripheral neurons connected to an active waist}}{n_{\text{waist}}^{\text{eff}}}. \tag{6}$$

If k neurons contribute equally and the rest not at all, the numerator is $(kP)^2$ and the denominator kP^2, so that $n_{\text{waist}}^{\text{eff}} = k$ as required. If one neuron carries everything, both sums are dominated by that neuron and $n_{\text{waist}}^{\text{eff}}$ tends to one. Section 4.4 shows what difference the correction makes to the ranking of waists in practice.

3.3 Macro-Scale Analysis: the τ-Core

3.3.1 Sources, Targets and Routing

The sources S are taken to be the sensory neurons, of which there are 12,711, and the targets T the motor and descending neurons, of which there are 1,382. Computing path centrality then requires a decision about which paths are to be counted, and this decision matters more than it might appear.

Two routing schemes are used. Under `levels` routing, a path may only traverse edges that do not decrease the hierarchical level, which makes the routing structurally acyclic and therefore well behaved, at the cost of discarding every lateral edge. Under `score` routing, paths follow a continuous flow score which recovers part of the lateral connectivity, with an explicit bound imposed on the length of admissible paths. Since almost 39% of all edges in this connectome are lateral, as Section 4.1 reports, the difference between the two schemes is substantial and both are reported throughout.

3.3.2 Why the Path Length Bound Is Mandatory

In a cyclic graph the number of source-to-target paths diverges combinatorially with the permitted path length, since paths may wander through cycles arbitrarily many times. As the count diverges, the ordering of nodes by path centrality degenerates, because a single well-connected neuron appears on an overwhelming proportion of the increasingly baroque paths. Section 4.2 quantifies exactly how bad this becomes. The bound on path length is therefore not a computational convenience that could be relaxed given more resources, but a definitional requirement without which the measure has no content.

3.4 Statistical Validation

3.4.1 Three Complementary Null Models

A single null model tests a single confound, and can therefore only ever establish that the observed structure is not explained by that one thing. Because several different properties of the connectome could plausibly generate an apparent excess of bow-ties, we use three null models, each of which destroys a different aspect of the observed structure while preserving the others.

The first, which we call model A, is the degree-preserving null of Milo et al. [9, 8]. It rewires the network through repeated edge swaps that leave the in-degree and out-degree of every neuron unchanged. It therefore preserves the degree sequences exactly and destroys the anatomy completely, so that a pattern which remains over-represented under it cannot be explained by degree alone. The second, model B, preserves regional structure instead: edges are resampled independently within each pair of anatomical groups at the observed density of that block, so that the connection density between any two regions is retained while individual degrees are destroyed. The third, model C, is the strictest of the three. Within each block it permutes the assignment of degrees to individual neurons, thereby preserving both the number of edges and the per-block degree sequences, and randomising only *which* waist neuron receives which degree. Model C consequently isolates the coincidence structure at the waist from everything else, which is exactly the property a bow-tie count depends on.

3.4.2 Inference

Significance is assessed with empirical p-values,

$$p = \frac{1 + \#\{c_{\text{rand}} \geq c_{\text{real}}\}}{N + 1}, \tag{7}$$

where c_{real} is the observed count and the c_{rand} are the counts obtained on N randomised networks. We use empirical p-values rather than z-scores because the null distribution of motif counts is strongly right-skewed, so that a z-score does not correspond to a tail probability and would systematically overstate significance.

The set of patterns to be tested is *pre-registered*: it consists of the twenty most frequent patterns of each of the sixteen signatures, three hundred and twenty patterns in all, selected before any randomisation was examined. This matters because selecting patterns after seeing which ones look unusual would guarantee significance by construction. Multiple testing across the three hundred and twenty tests is controlled with the Benjamini–Hochberg false discovery rate procedure [2], which bounds the expected proportion of false positives among the rejections. Randomisation uses $N = 100$ samples with a fixed seed, and the mixing of the edge-swap chain was verified empirically rather than assumed by checking that the results stabilise well before the number of swaps actually used.

3.4.3 Effect Size

The ratio between the real count and the median randomised count is deliberately *not* reported as an effect size anywhere in this thesis. It spans several orders of magnitude across the three null models and is dominated by how thoroughly each model destroys the structure rather than by any property of the data, which makes it uninterpretable as a measure of biological strength. Section 4.5.1 shows that the ratio does not even preserve the ordering between signatures from one null to another, which settles the matter.

3.5 Computational Realisation

3.5.1 The Matrix Engine

A direct implementation of Equation (5) loops over combinations of groups and performs, for each one, a copy followed by k_{out} multiplications and as many additions over arrays of the length of the waist. On the shallow window this amounts to roughly 1.7 billion iterations, and profiling shows something initially surprising: the cost is dominated not by the arithmetic but by the per-call overhead of the array library, some two microseconds per call. The computation is therefore limited by how often the inner loop crosses into the numerical library, not by how much numerical work it does.

The remedy is to arrange for all combinations of a given waist to be computed at once. Let P_{in} be the matrix whose row r is the element-wise product of the input flow vectors of combination r, and let P_{out} be the analogous matrix for the output combinations. Then every count belonging to that waist is an entry of

$$T = P_{\text{in}} P_{\text{out}}^{\top}, \tag{8}$$

a single dense matrix product which can be delegated wholesale to the optimised linear algebra library. Billions of individual calls collapse into a few thousand matrix products.

The compression metrics of Section 3.2.3 follow the same pattern and need not be computed separately per combination. Since the per-combination vector is the element-wise product of an input row and an output row, the sum of its squares and the count of its non-zero entries are themselves matrix products,

$$\textstyle\sum_i p_{v,i}^2 = P_{\text{in}}^2 (P_{\text{out}}^2)^{\top}, \qquad n_{\text{active}} = (P_{\text{in}} > 0)(P_{\text{out}} > 0)^{\top}, \tag{9}$$

so that the vector itself is never materialised for any individual combination.

3.5.2 The Sparse Engine

Equation (8) requires memory proportional to the number of combinations multiplied by the size of the waist, and therefore switches itself off precisely on the largest waists. This is unfortunate, because those are the expensive ones: on the shallow window the waists that exceed the memory budget account for the majority of the total runtime.

The way out comes from observing that on a large waist the flow vectors are almost entirely zero. An input group typically reaches only a few hundred of the 24,835 neurons of the largest metanode, so the matrices in Equation (8) are dense representations of almost nothing. Computing the same product with sparse operands produces directly, and only, the pairs whose count is non-zero, and those are exactly the pairs that need to be emitted. Memory then scales with the number of non-zero entries rather than with the product of the dimensions, and the constraint that forced the fallback disappears. The three quantities computed alongside the count share the same sparsity structure, since the values are non-negative counts and a product vanishes if and only if one of its factors does; that this alignment holds is asserted at run time rather than assumed.

3.5.3 Output Distillation

Exhaustive counting produces one row of output per pattern, and for the full window that means 512,969,374 rows. Retained as plain text these rows occupy more than 100 GB across the four windows, which is unwieldy to store and slow to read.

Examining what the downstream analyses actually consume shows that none of them uses the individual rows. Every one of them either aggregates the rows into summary counts or selects a top-N subset by frequency. The row-level output is therefore an intermediate rather than a result: it needs to exist for the duration of a single pass and no longer. The search engine consequently retains only the aggregates, computed by signature, by waist and by level chain, together with top-N selections in which the complete rows are kept. This reduces the four windows from over 100 GB to 68 MB while producing downstream results that are identical.

The selections are deliberately wider than any current analysis requires, and they are stratified by waist and signature jointly as well as by waist alone, so that comparisons made at parity of waist between classes of motif remain possible. Section 4.7 depends on exactly that stratification.

3.6 Correctness Verification

Three independent mechanisms are used to establish correctness, and they are described here in some detail because the credibility of every number in Section 4 rests on them.

The first is a reference implementation. A slow but readable implementation of Equation (5), written directly from the mathematics without any of the optimisations, is retained alongside the fast engine. The two are compared row by row across all 1,043,872 rows and all 26 columns of a complete run, including 24 waists deliberately forced onto the sparse path by lowering the memory budget so that the sparse code is exercised rather than merely present.

The second is exhaustive enumeration. Every closed-form count in this thesis is checked against explicit enumeration on random graphs where enumeration is feasible: the within-area formula of Section 4.9 on three hundred random graphs, the symmetric-polynomial formulation used for the $(k_{\text{in}}, k_{\text{out}})$ sweep, and the pattern algebra of Proposition 3.1. A closed form that agrees with brute force on small cases can still be wrong on large ones, but a closed form that disagrees on small cases is certainly wrong, and this check has caught real errors during development.

The third is internal consistency. Each analysis recomputes the occurrence count of every pattern it handles directly from the graph and compares it against the stored value, so that a corrupted or misaligned intermediate would be detected at the point of use. All such checks reported exact agreement.

4 Results

4.1 Hierarchical Structure of the Connectome

4.1.1 Level Sizes and Composition

Since the level assignment is inherited rather than derived here, the first thing to establish is that it means what it is supposed to mean. Table 2 reports how the neurons are distributed across the five levels together with the functional composition of each level.

Table 2: Neurons per hierarchical level and functional super-class composition (percentages by row).

Level	Neurons	%	optic	sensory	central	descending	vis. proj.
L1	13,601	10.1	51.7	**42.3**	3.9	0.0	0.6
L2	47,049	35.1	75.6	8.6	11.6	0.1	2.3
L3	44,265	33.0	67.7	3.2	21.3	0.3	6.1
L4	17,124	12.8	26.9	4.9	48.1	1.4	16.3
L5	12,142	9.0	3.3	5.5	**72.0**	**7.0**	8.4

The gradient is monotone in the expected direction. Sensory content falls from 42.3% at level 1 to 5.5% at level 5, central content rises from 3.9% to 72.0% over the same range, and descending neurons, which carry signals out of the brain towards the motor system, appear only at the two deepest levels. None of this was imposed by the level assignment procedure, which did not use the super-class labels, so the agreement constitutes independent confirmation that the hierarchy captures a genuine sensory-to-motor axis rather than an arbitrary partition of the neurons.

4.1.2 Connection Density

Raw counts of edges between levels are dominated by the sizes of the levels involved: level 2 alone contains 47,049 neurons, so any transition touching it appears large simply on that account. The informative quantity is therefore the density, defined as the number of edges observed per ten thousand pairs of neurons that could possibly be connected. Figure 2 shows the full matrix.

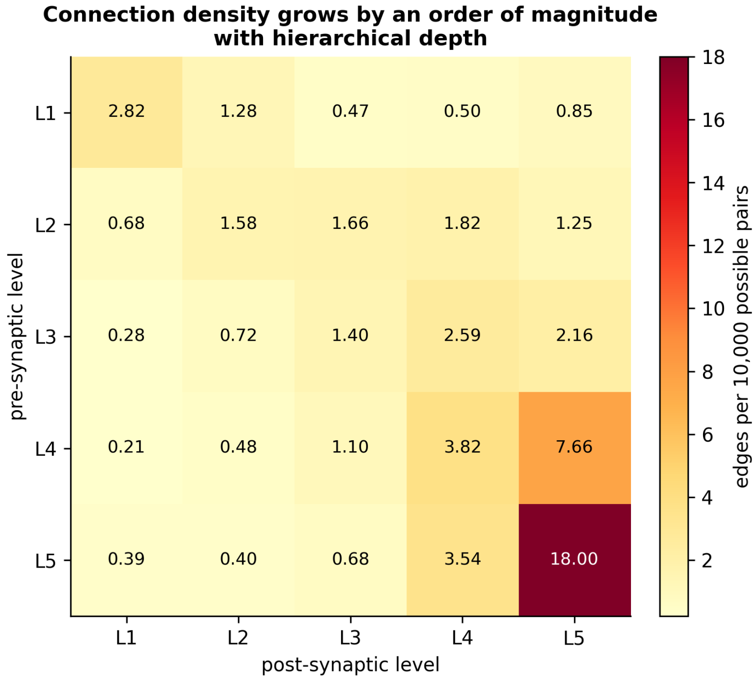

Figure 2: Connection density between hierarchical levels, expressed as edges per ten thousand possible pairs. The normalisation removes the effect of level size, which dominates raw edge counts. Density grows by an order of magnitude with depth, from 1.28 along L1→L2 to 18.00 within L5.

Density grows by an order of magnitude with depth, from 1.28 along the L1-to-L2 transition to 18.00 within level 5 itself. The deep brain is therefore not a smaller copy of the shallow brain but a qualitatively denser structure, in which each neuron participates in far more connections relative to the population available to it. This single observation recurs throughout the results and explains several of them.

4.1.3 Edge Direction

Classifying every edge of the connectome by Equation (2) gives 1,169,787 feed-forward edges, 1,054,005 lateral edges and 476,721 feedback edges. Almost 39% of all connections are therefore lateral, running between neurons of the same hierarchical level. This has an immediate methodological consequence: any routing scheme that discards lateral edges is discarding a very substantial fraction of the connectome, which is why Section 4.2 reports both schemes rather than choosing one.

4.2 The Macro-Scale Hourglass

Table 3 reports the hourglass score under both routing schemes and several bounds on path length, all at $\tau = 0.9$.

Table 3: Hourglass score under different routing schemes and path-length bounds, at $\tau = 0.9$. The last row is degenerate and is reported to show why the bound is mandatory.

Routing	Max hops	$S \to T$ paths	$\lvert C(\tau)\rvert$	$\lvert C_f(\tau)\rvert$	H
`levels`	4 (structural)	1,995,460	154	307	**0.498**
`score`	3	4,236,294	331	456	0.274
`score`	4	1.20×10^8	326	433	0.247
`score`	5	3.67×10^9	269	401	0.329
`score`	∞	1.17×10^{75}	1	10	*(degenerate)*

Under structural routing the score is 0.498, which is to say that the real core is half the size that the dependency structure alone would require. By the definition of Equation (4) the hourglass property is therefore present, and Figures 3 and 4 show that the finding does not hinge on the particular threshold chosen.

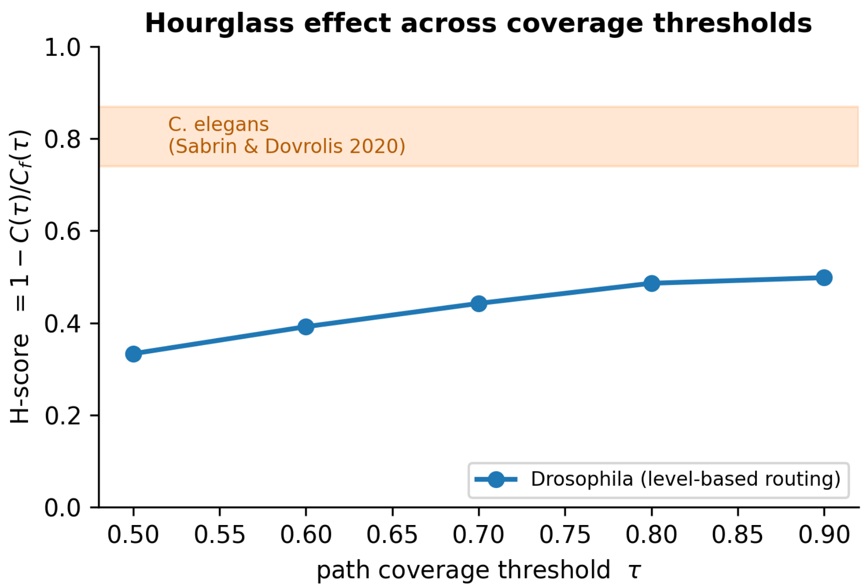

Figure 3: Hourglass score as a function of the coverage threshold τ, under level-based routing. The score is stable over a broad range of τ, which is what makes the single value reported at $\tau = 0.9$ meaningful rather than an artefact of one threshold.

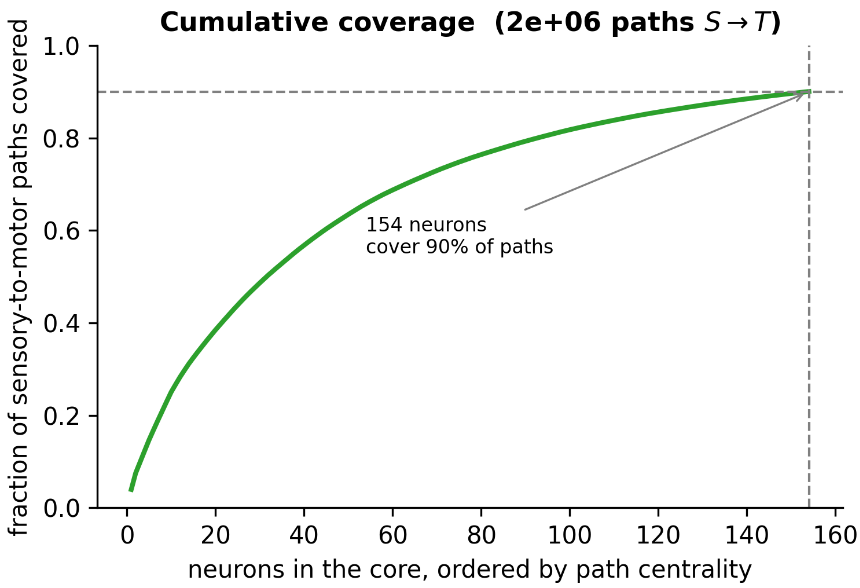

Figure 4: Cumulative fraction of sensory-to-motor paths covered as neurons are added to the core in decreasing order of path centrality. The sharp initial rise is the hourglass property in its rawest form: 154 neurons out of 134,181 carry 90% of the end-to-end flow.

The score is nevertheless routing-dependent, ranging from 0.247 to 0.498 across the schemes tested, and the honest way to report it is therefore the full table rather than any single figure drawn from it. The final row of the table shows what happens when the bound on path length is removed altogether: the count of source-to-target paths reaches 1.17×10^{75}, a single neuron then covers 97.6% of them, and the τ-core collapses to one element. At that point the measure has lost all content, which is the concrete justification for the requirement stated in Section 3.3.

That the observed concentration is not an artefact of the degree sequence was checked directly against randomised networks, as Figure 5 shows.

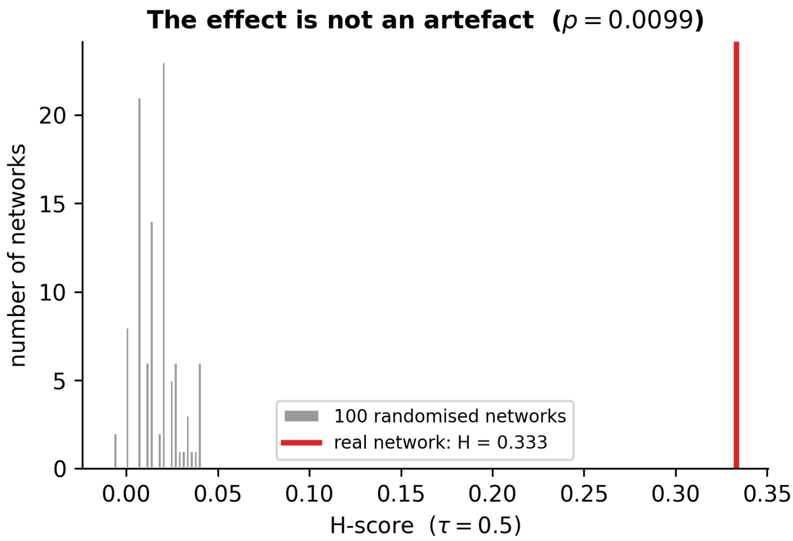

Figure 5: Distribution of the hourglass score over randomised networks, against the observed value. The observed score falls well outside the null distribution, establishing that the concentration of flow is not a by-product of the degree sequence alone.

4.2.1 Weaker Than in *C. elegans*, and Why That May Be Expected

The reference study of Sabrin et al. [10] reports hourglass scores between 0.74 and 0.87 for the *C. elegans* connectome. The values obtained here, between 0.25 and 0.50, are substantially lower. The *Drosophila* brain is therefore hierarchically organised, but it funnels its flow through a narrow waist considerably less than the nematode does, and the difference is large enough to require an explanation rather than a remark.

A plausible one, offered here as an argument to be weighed rather than as a demonstrated result, is that the two nervous systems are solving different problems. *C. elegans* routes essentially all of its sensory information through ten to fifteen command interneurons, which constitutes a single shared bottleneck for the whole animal. The fly processes several sensory modalities in parallel, and each of them plausibly has its own bottleneck rather than sharing one. The composition of the τ-core supports this reading, since it separates into distinct clusters, with `AL` structures serving olfaction, `GNG` and `PRW` serving taste and mechanosensation, and `LH.SLP` forming a third group, rather than forming a single shared core. Several parallel hourglasses will necessarily produce a lower global score than one deep hourglass, even when each individual pathway is strongly funnelled, because the union of several small cores is larger than any one of them.

4.3 Bow-Tie Motifs Across Level Windows

4.3.1 The Four Windows

Table 4 summarises the exhaustive search over all four level windows, which to our knowledge has not previously been carried out on this connectome.

Table 4: Exhaustive bow-tie motif search across level windows, with $k_{\text{in}} = k_{\text{out}} = 2$, $max\_jump = 1$ and $min\_count = 10$.

Window	Neurons	Metanodes	Motifs	Runtime
$\{1, 2, 3\}$	104,915	879	31,184,800	14m39s
$\{2, 3, 4\}$	108,438	1155	119,026,375	57m43s
$\{3, 4, 5\}$	73,531	1220	433,546,629	3h05m51s
$\{1, 2, 3, 4, 5\}$	134,181	—	512,969,374	3h34m49s

The motif count grows steeply with depth, from thirty-one million through one hundred and nineteen million to four hundred and thirty-four million, while the number of neurons does not grow at all: the deepest window contains fewer neurons than the shallowest. Figure 6 makes the dissociation visible. Growth is therefore driven by connection density rather than by the size of the search space, exactly as the density measurements of Section 4.1 would predict.

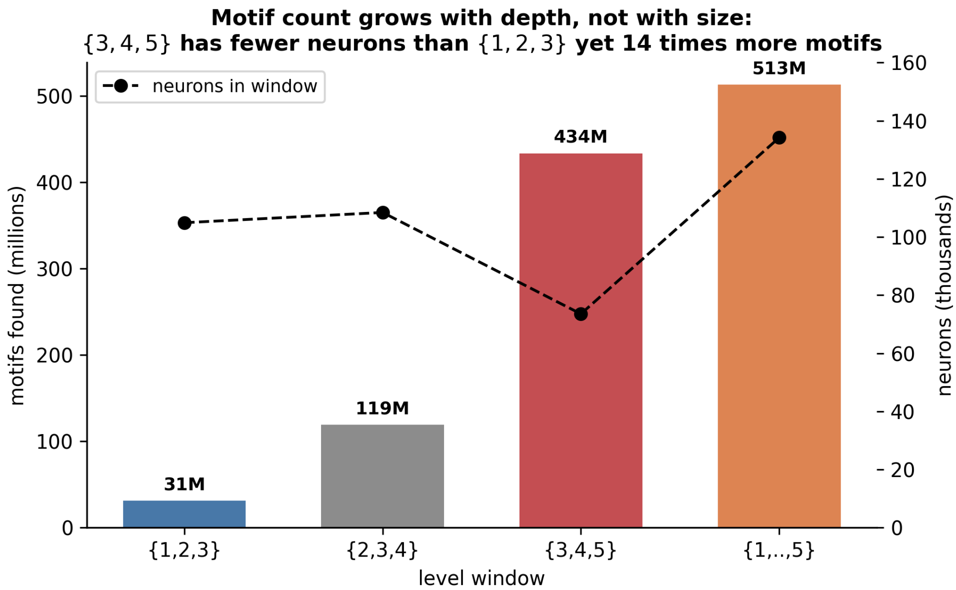

Figure 6: Number of bow-tie motifs found in each level window, shown as bars against the left axis, together with the number of neurons the window contains, shown as a dashed line against the right axis. The two do not track each other: window $\{3, 4, 5\}$ contains fewer neurons than $\{1, 2, 3\}$ yet yields fourteen times as many motifs.

The scale of the pruning described in Section 3.2 can be read off the shallow window, where the search examined 8,310,710,163 candidate combinations. Of these, 112,688,248 were excluded by the disjointness constraint and 8,125,079,064 were discarded because one of their flow factors was identically zero, leaving 72,942,851 raw motifs of which 31,184,800 passed the minimum count threshold. More than ninety-seven per cent of the nominal space is thus eliminated without any loss of exactness.

4.3.2 Directional Signatures

The coarse four-way classification turns out to be nearly vacuous. On the shallow window, *mixed* accounts for 29,453,022 patterns, which is 94.4% of the total, while *pure_lateral* accounts for 1,651,892 (5.3%), *pure_FF* for 57,821 (0.19%) and *pure_FB* for only 22,065 (0.07%). This is not a surprising finding so much as an arithmetic one: with four edges each independently able to run in three directions, complete agreement among all four is simply improbable, and the coarse label therefore carries very little information.

The sixteen-signature decomposition of Equation (3) resolves the *mixed* bucket into thirteen distinct classes whose behaviour differs considerably, as Figure 7 shows.

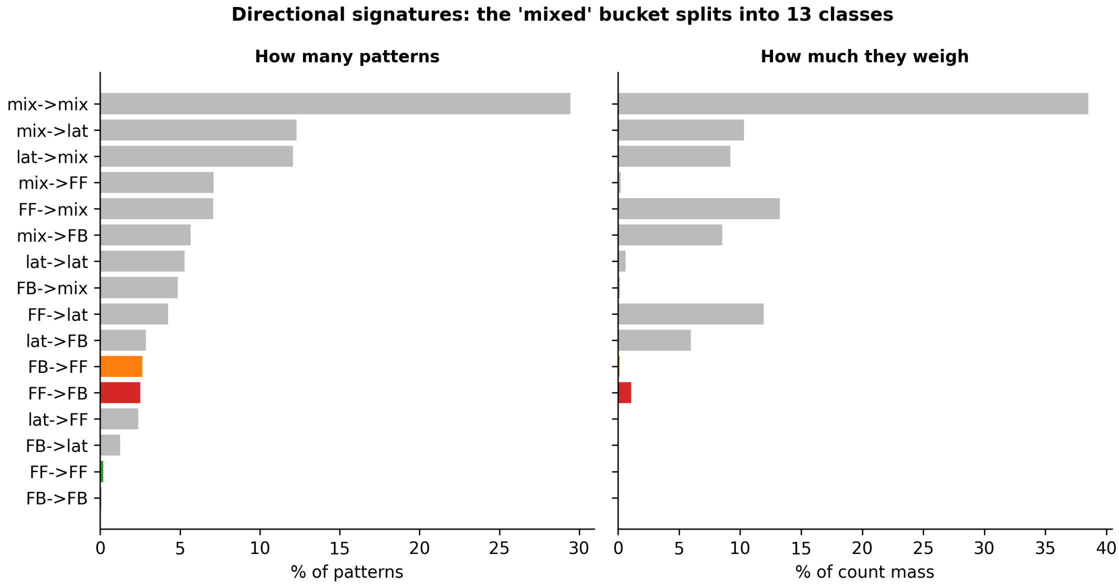

Figure 7: Decomposition of the sixteen directional signatures on window $\{1, 2, 3\}$, by number of patterns on the left and by share of total count mass on the right. The two panels disagree substantially: `mix->FF` is frequent but carries almost no mass, while `FF->mix` is the reverse. The pure signatures `FF->FF` and `FB->FB` are highlighted, and their rarity is what makes the coarse four-way classification uninformative.

The two panels of that figure disagree in an instructive way. A signature may account for many distinct patterns while contributing almost nothing to the total count, as `mix->FF` does, or account for relatively few patterns each of which is realised enormously often, as `FF->mix` does. Reporting only one of the two quantities would therefore give a misleading picture, and both are reported throughout.

Comparing the shallow and deep windows reveals that this composition is not a fixed property of the connectome but changes substantially with depth.

Table 5: Share of total motif mass by directional signature, shallow versus deep window.

Signature	$\{1, 2, 3\}$ (% mass)	$\{3, 4, 5\}$ (% mass)
`lat->mix`	9.2	**42.7**
`mix->lat`	10.3	**23.3**
`lat->lat`	0.6	**10.2**
`mix->mix`	**38.5**	13.9
`FF->mix`	13.3	1.5
any purely lateral branch	*20.1*	***76.2***

Signatures containing at least one purely lateral branch carry 20.1% of the mass in the shallow window and 76.2% in the deep one, while `mix->mix`, which dominates the shallow window at 38.5%, falls to 13.9%. Figure 8 displays the same comparison signature by signature, sorted by the magnitude of the shift. The deep brain is, in this precise sense, laterally dominated: its characteristic circuit is one in which information enters or leaves a waist by moving sideways within a processing stage rather than by advancing through the hierarchy.

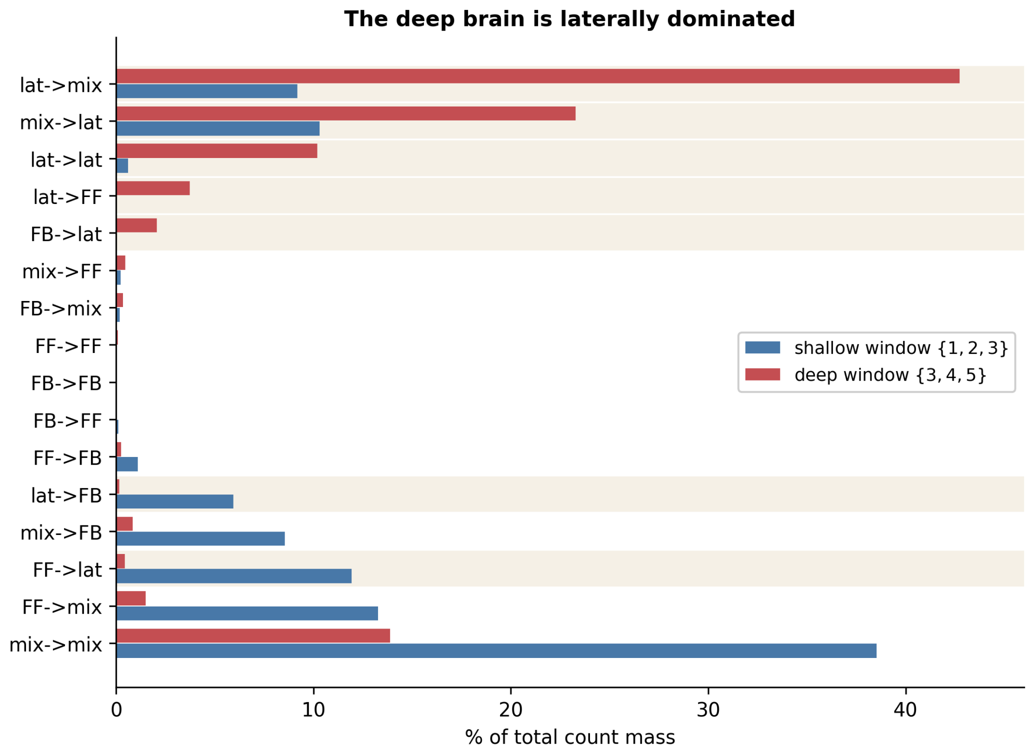

Figure 8: Share of total count mass by directional signature in the shallow and deep windows, sorted by the size of the shift between them. Shaded rows are signatures containing a purely lateral branch, and they gain almost exactly what the others lose. `lat->mix` alone rises from 9.2% to 42.7%.

The highest-count individual patterns illustrate the same point concretely. Figure 9 shows the single most frequent bow-tie of the shallow window, and Figures 10 to 13 give representative examples of every signature class.

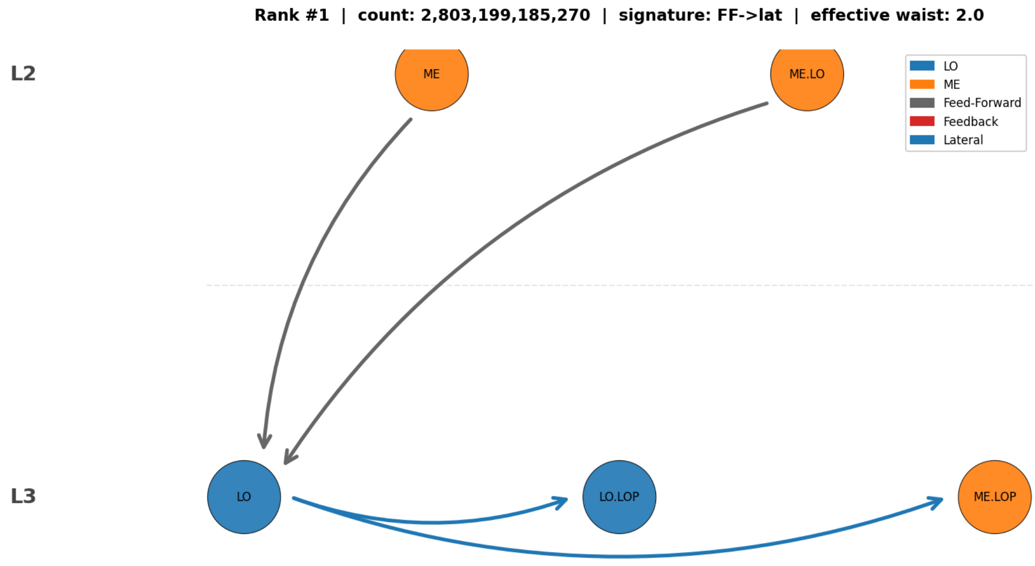

Figure 9: The single most frequent bow-tie motif of window $\{1, 2, 3\}$, with 2.80×10^{12} instances and signature `FF->lat`. Two medulla groups at level 2 converge onto the lobula at level 3, which then distributes laterally within level 3. Grey edges are feed-forward and blue edges lateral. The pattern is a compact illustration of the general finding: the dominant structures combine a feed-forward entry with a lateral spread rather than being purely feed-forward.

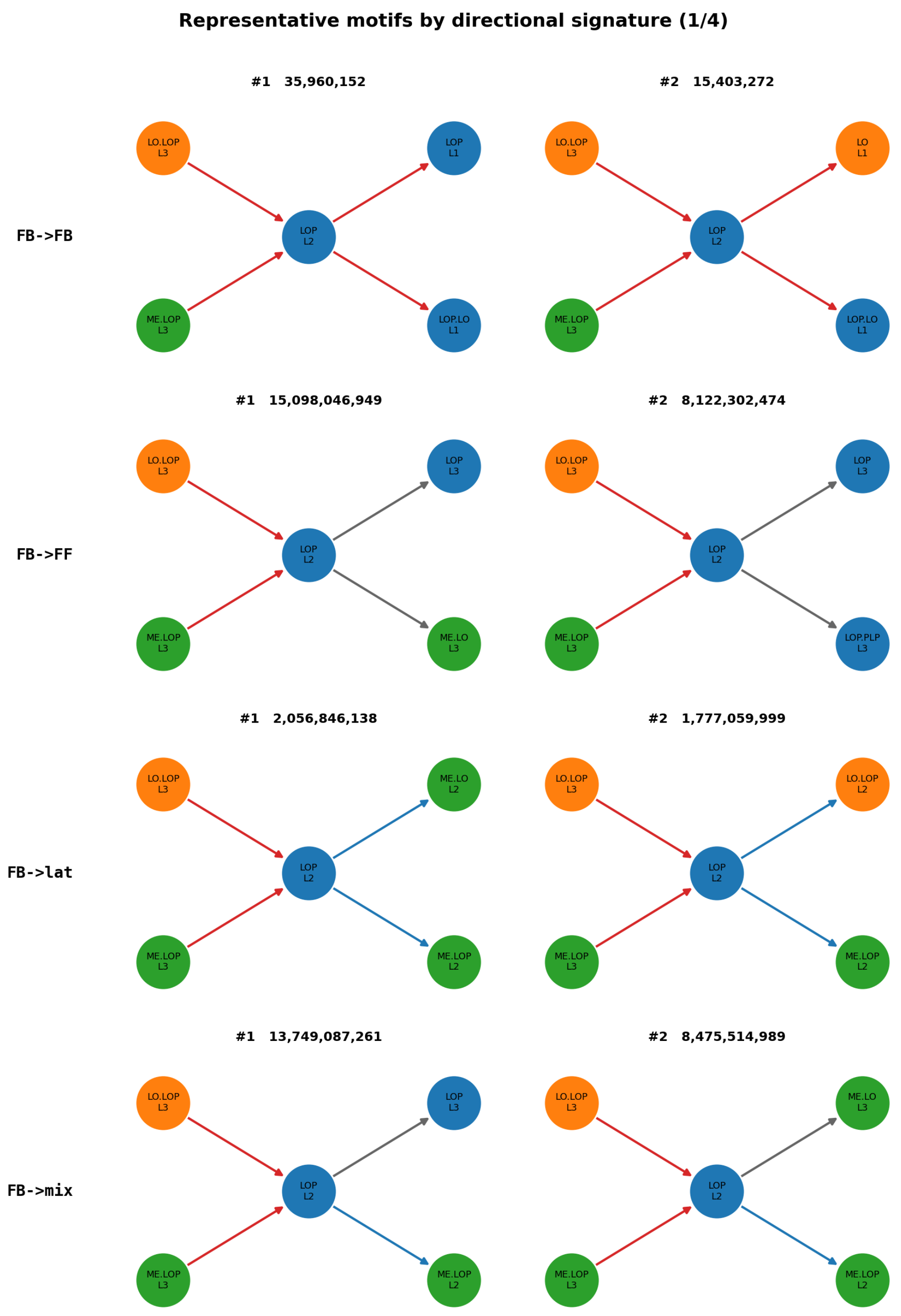

Figure 10: Representative bow-tie motifs for the first four directional signatures, window $\{1, 2, 3\}$. Each panel places the input groups on the left, the waist in the centre and the output groups on the right; node labels give the anatomical group and its hierarchical level, node colour encodes the macro-area, and edge colour the direction, with grey for feed-forward, red for feedback and blue for lateral. The counts above each panel are numbers of neuron-level instances, and they differ between classes by nearly three orders of magnitude.

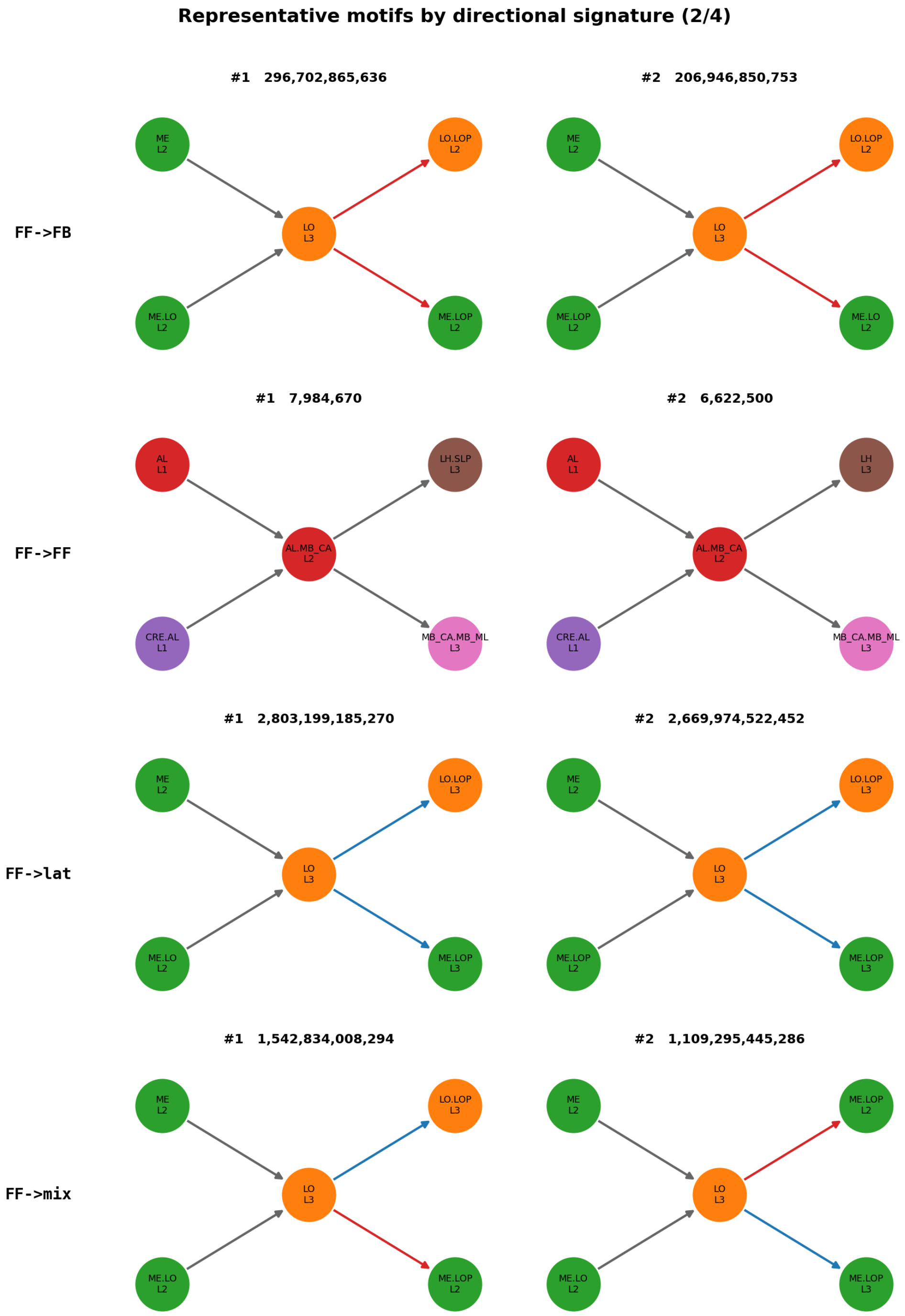

Figure 11: Representative motifs, signatures five to eight, with conventions as in Figure 10. The optic-lobe groups dominate the highest-count patterns throughout, consistent with the density reported in Section 4.1.

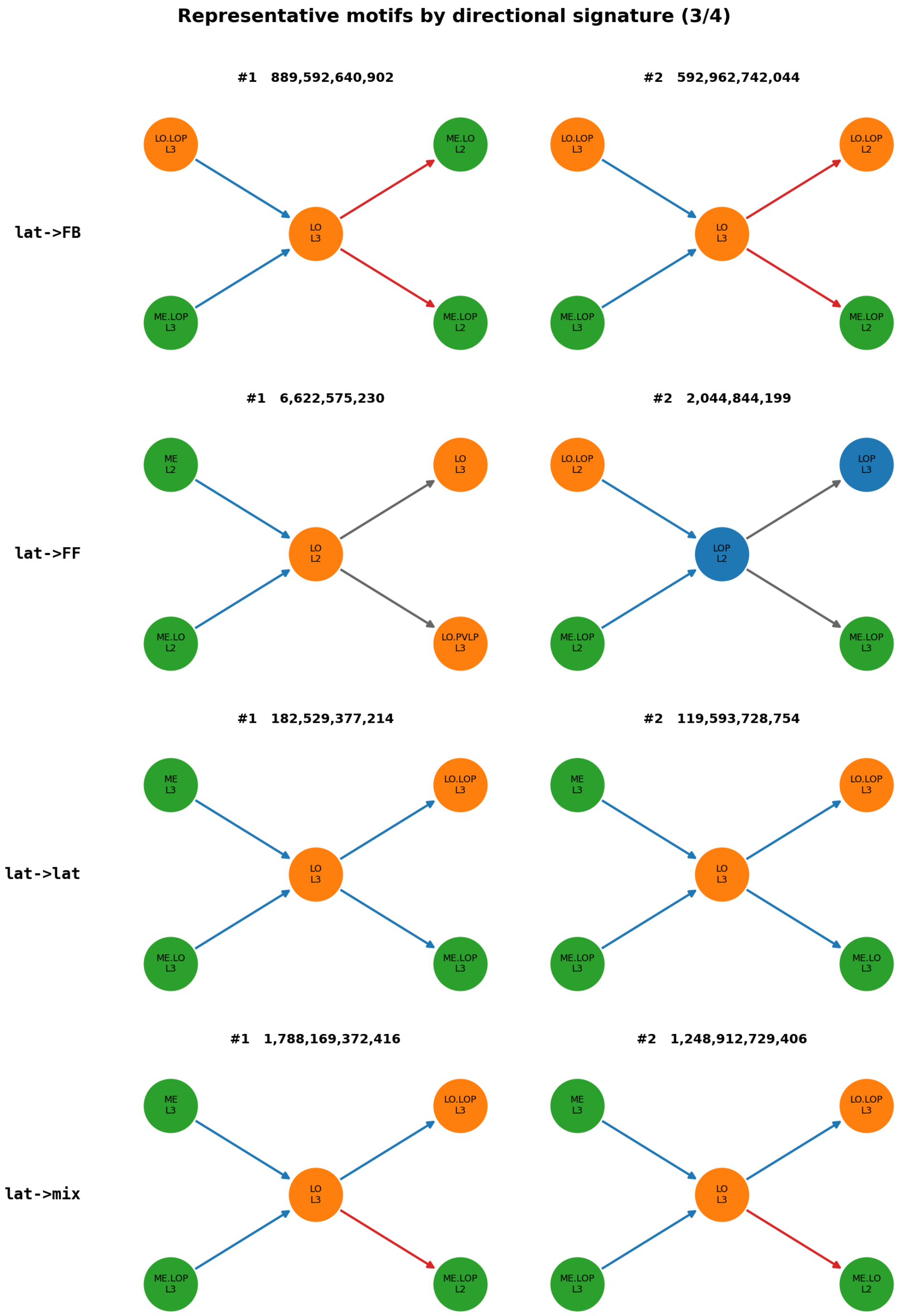

Figure 12: Representative motifs, signatures nine to twelve, with conventions as in Figure 10.

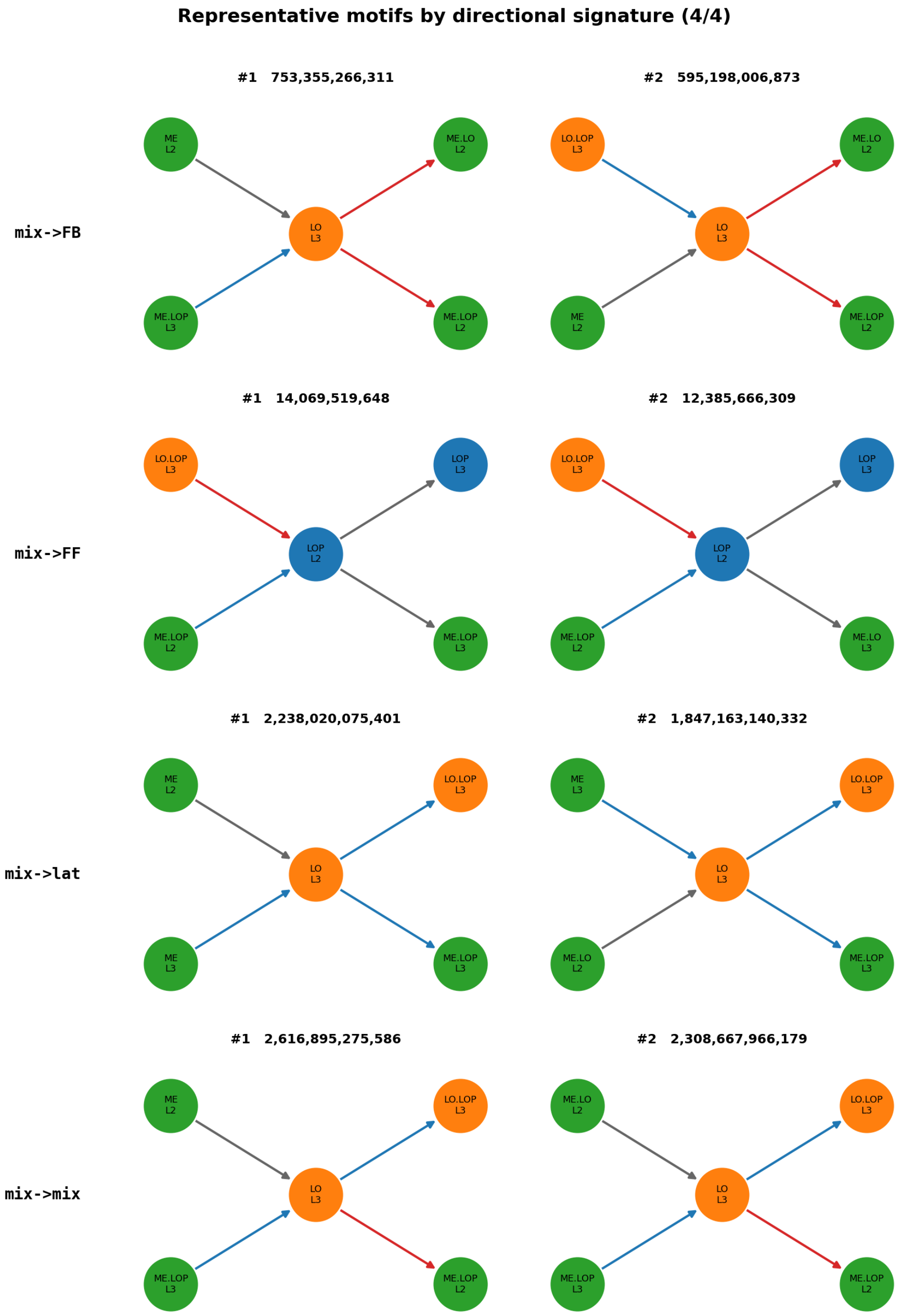

Figure 13: Representative motifs, the remaining signatures, with conventions as in Figure 10.

4.3.3 Level Chains

A complementary view attributes the mass of each pattern to the triple consisting of the level of its inputs, the level of its waist and the level of its outputs. This is coarser than the signature, since it records only which levels are involved rather than how the branches are oriented, but it answers a question the signature cannot: where in the hierarchy the bottlenecks actually sit.

In the deep window the dominant chains are L3 to L3 to L3 at 45.6% of the mass, L3 to L3 to L4 at 28.0%, L4 to L3 to L3 at 16.6% and L4 to L3 to L4 at 3.3%, as Figure 14 shows.

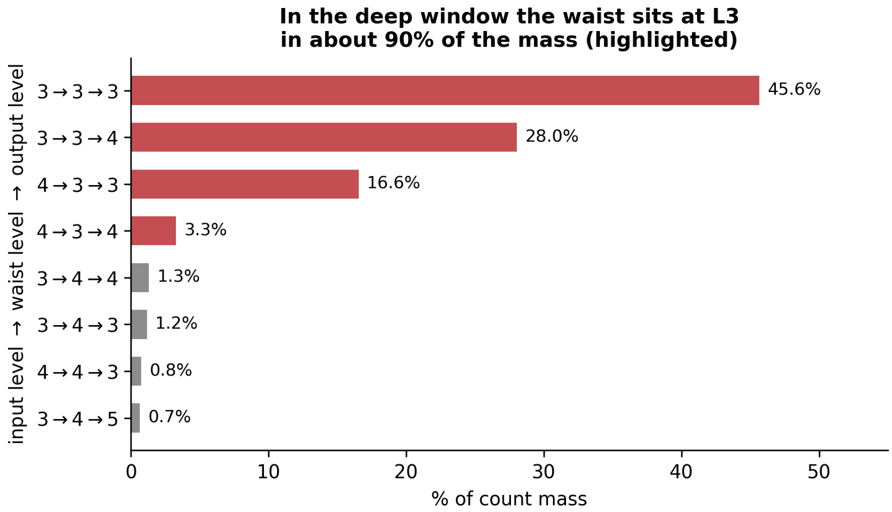

Figure 14: Distribution of motif mass across level chains in the deep window, where a chain records the input level, the waist level and the output level. Chains whose waist lies at level 3 are highlighted, and together they account for roughly 90% of the mass.

The waist sits at level 3 in roughly ninety per cent of the mass, which is worth pausing on. Level 5 is the most internally dense level of the entire connectome, as Section 4.1 established, and yet it barely appears as a waist at all. High internal density combined with very few bottlenecks is a specific structural signature: it indicates connectivity that is recurrent and lateral rather than convergent and divergent. A region can be densely wired without any of its neurons functioning as an obligatory passage point, and level 5 appears to be exactly such a region. That distinction is invisible to density measurements taken on their own, and it is one of the clearer instances in this work of the two scales of analysis saying different things about the same tissue.

4.3.4 The Shallow Window, and a Direct Answer About L1

The same analysis on the shallow window gives a sharper picture still, in which four chains account for 99.2% of the entire count mass.

Table 6: Level chains in the shallow window: four chains carry almost all the mass, and all four have their waist at level 3.

Chain	Share of mass
L2 → L3 → L3	33.3%
L3 → L3 → L2	24.4%
L2 → L3 → L2	21.6%
L3 → L3 → L3	19.9%
total	**99.2%**

All four have their waist at level 3, which is the same concentration observed independently in the deep window and therefore not a peculiarity of either.

The table also answers directly a question that motivated this part of the analysis, namely how many bow-tie patterns run from level 1 into level 2. The answer is that almost none do. Every chain involving level 1 together accounts for roughly 0.02% of the mass, despite level 1 containing 13,601 neurons. The explanation is functional rather than an artefact of the method: level 1 is 94% photoreceptors and sensory neurons, which receive very little from within the brain and project forward. They are sources of flow rather than points of convergence, and a bow-tie requires convergence. A neuron population can therefore be large and still be a poor waist, which is a useful corrective to the intuition that the biggest structures must be the most important ones.

4.4 Waist Compression

The nominal compression metric fails in exactly the way anticipated in Section 3.2.3. Its ranking is headed by `LO.PVLP_L2`, which attains a nominal compression of 2431 with only *two* active neurons out of twenty-one, while the biologically prominent `AL.MB_CA_L2` scores merely 40. The first of these is not a bottleneck in any meaningful sense; it is a pair of neurons in a group that happens to be large.

Under the realised metric of Equation (6) the ranking becomes interpretable. On the shallow window `AL.MB_CA_L2` has 144 neurons in its group, of which 118 are active, an effective count of 4.8, and 2694 peripheral neurons connected through it, giving a realised compression of 561.8. The metric now reflects the structure one wants it to reflect, namely how much of the periphery is funnelled through how many genuinely participating neurons.

On the deep window the analysis retains 19,416 patterns, of which 794 have an effective waist of at least five neurons, and the internal consistency check described in Section 3.6 reproduced all 19,416 counts exactly. The leading structure there is `ME_L4`, with a realised compression of 891.8 over an effective waist of 5.7 neurons, followed by several variants of `ME.LO_L3`: the optic lobe dominates compression in the deep brain as it dominates the motif counts. Figure 15 plots the whole distribution.

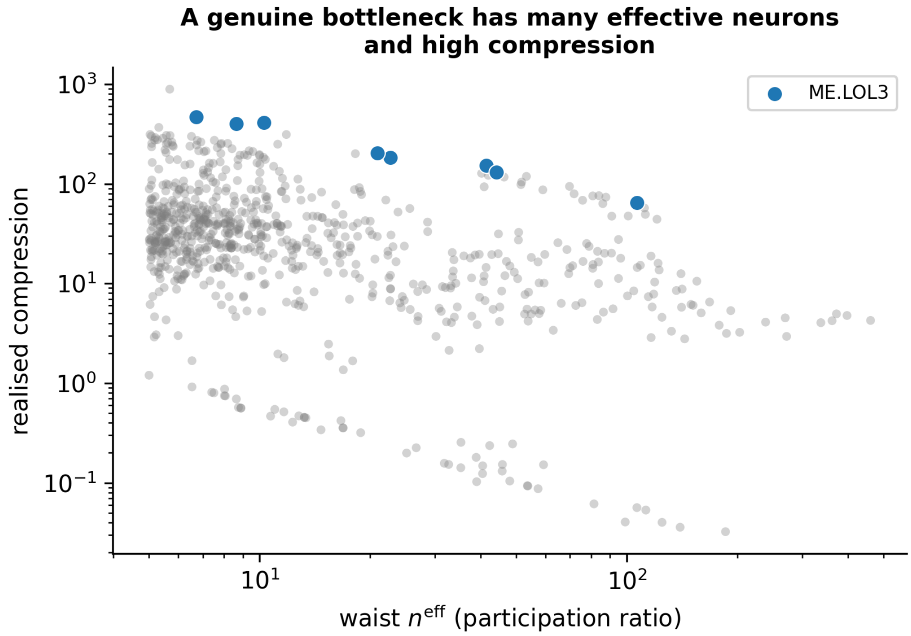

Figure 15: Realised compression against the effective number of waist neurons for window $\{3, 4, 5\}$, with the `ME.LO_L3` family highlighted. Points at the far left achieve high compression through a single active neuron and are precisely the artefacts that the participation ratio is designed to expose. A genuine bottleneck lies towards the upper right, having both many effective neurons and high compression.

4.4.1 Two Anatomically Distinct Forms of Recurrence

Restricting attention to the two recurrent signatures reveals something that the aggregate figures conceal, namely that the two do not live in the same place.

The re-entrant form, `FF->FB`, comprises 786,216 patterns, which is 2.52% of the total, carrying a mass of 1.12×10^{12} or 1.10%. Its waist is the lobula, `LO_L3`, in 88.5% of that mass, and its leading pattern has two medulla groups converging onto the lobula which then projects onto two lobula plate groups. The relay form, `FB->FF`, comprises a very similar number of patterns, 822,466, but carries nine times less mass, and it sits on a different structure altogether: the lobula plate, `LOP_L2`, in 93.2% of its mass. Pure feedback, `FB->FB`, is negligible in mass and also lives on `LOP_L2`.

The two forms of recurrence are therefore anatomically separated rather than being two descriptions of one phenomenon. Re-entry into the lobula and relay through the lobula plate are distinct circuits, and both belong to the optic lobe, which dominates the recurrent structure of this window as it dominates much else.

This dominance survives the correction of the compression metric, but it must be qualified in one respect. Under realised compression the top of the ranking remains optic, with `ME.LO_L3` at 2173, `ME.LO_L2` at 796 and `ME_L3` at 690, but now with substantial counts, between 4×10^5 and 1.4×10^7, and with genuine participation rather than with two active neurons out of twenty-one. The dominance is thus a property of the circuitry rather than an artefact of group size, which is a stronger claim than the nominal metric could have supported.

4.5 Statistical Significance

Of the three hundred and twenty pre-registered patterns, all three hundred and twenty had their observed counts reproduced exactly from the graph before any testing began. After Benjamini–Hochberg correction at $q < 0.05$, significance was obtained for 319 of 320 patterns under model A and for 310 of 320 under both models B and C.

That the strictest model yields nearly the same number of significant patterns as the weakest is the informative outcome. Model C preserves both the block structure and the per-block degree sequences and randomises only the alignment between them, so a pattern that remains significant under it cannot be explained by degree sequences, nor by regional connection densities, nor by which particular waist neurons happen to carry which degrees. The convergence–divergence structure of these bow-ties is therefore not accounted for by any of the three confounds tested.

4.5.1 The Ranking Between Signatures Inverts Between Null Models

The counts of significant patterns are similar across the three models, but the *ordering* of signatures by degree of enrichment is not merely different between them: it inverts.

Table 7: Median ratio of real to randomised count, by signature, under two complementary null models. The extremes swap places.

Signature	Null A (degrees kept)	Null B/C (anatomy kept)
`FF->FF`	2.21×10^5 *(highest)*	8.4–11.4 *(lowest)*
`mix->FF`	2.06×10^3	4.7×10^4
`FB->FB`	1.98×10^3	9.2×10^3
`lat->lat`	1.84×10^3	6.4×10^5

The signature `FF->FF` is the most enriched of all when degree sequences are preserved and anatomy destroyed, and the *least* enriched when anatomy is preserved and the degree alignment destroyed. The signature `lat->lat` moves in the opposite direction by more than two orders of magnitude.

The interpretation is straightforward once stated. Purely feed-forward bow-ties are rare given the degree sequence alone, so a null model that ignores anatomy finds them strikingly over-represented in the real network; but once the regional block structure is preserved, they turn out to be close to what that structure already implies, because the anatomy itself is broadly feed-forward. Lateral bow-ties behave in exactly the opposite way: unremarkable given the degrees, strongly over-represented given the anatomy.

This is the strongest argument this thesis can offer for using several null models rather than one. A single null would not merely have given a weaker answer; depending on which one had been chosen, it would have supported the opposite ranking of signatures, with nothing internal to that analysis to indicate that anything was amiss.

4.6 Micro–Macro Cross-Validation

Comparing the two scales on the six hundred and sixty groups of the shallow window gives a Spearman rank correlation of $\rho = +0.121$ with $p = 0.0018$ between the motif mass of a group acting as a waist and its coverage in the τ-core. A rank correlation was used rather than a linear one because both quantities span several orders of magnitude and neither is normally distributed; what it measures is whether groups that rank highly on one scale tend to rank highly on the other. The value is small but reliably different from zero, which means there is a real signal and that it is far from a one-to-one relationship.

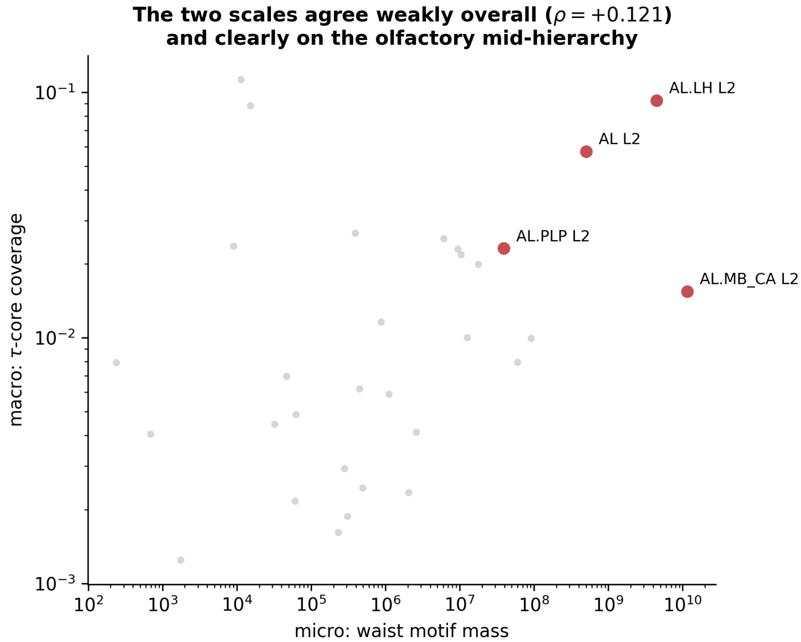

Figure 16: Waist motif mass at the local scale against τ-core coverage at the global scale, one point per group and level, with both axes logarithmic. The overall association is positive but weak. Highlighted are the mid-hierarchy groups of the olfactory pathway, on which the two scales agree closely.

Where the two scales agree, they agree on the mid-hierarchy waists of the olfactory pathway, as Table 8 shows.

Table 8: Groups where the two scales agree: mid-hierarchy waists of the olfactory pathway, with ranks out of six hundred and sixty groups.

Group	Micro rank (motif mass)	Macro rank (coverage)
`AL.MB_CA` L2	8	12
`AL.LH` L2	10	2
`AL` L2	20	4
`AL.PLP` L2	50	8

The overlap between the two top-twenty lists is three groups out of twenty, against 0.61 expected by chance under a hypergeometric model, giving $p = 0.0195$. That agreement is modest in absolute terms but is concentrated on structures of known functional importance, which makes it more meaningful than the raw number suggests.

The scales disagree on the input-side groups, and the disagreement is worth examining rather than dismissing. Groups such as `GNG` and `PRW` at level 1 are central in the global flow, appearing in the τ-core, while ranking very low as local waists. There is no contradiction here once the two measurements are read carefully: being an obligatory passage point on the way from sensation to action is a different property from being a point at which many streams converge and then diverge locally. A relay through which everything must pass will score highly on path centrality and poorly on bow-tie counts, and the fly's early sensory structures are precisely such relays.

On the deep window the global correlation is comparable, at $\rho = +0.103$ with $p = 0.001$, but the overlap of the top-twenty lists falls to one group out of twenty and is no longer significant, with $p = 0.33$. This is the expected outcome rather than a failure of replication, since the τ-core is dominated by olfactory and gustatory pathways which reside at levels one to three and are therefore largely outside the deep window.

4.7 Neurotransmitters: A Negative Result

4.7.1 What Holds

Before turning to what does not hold, it is worth stating what does. The composition of excitatory and inhibitory neurons by anatomical group is robust across every window examined: `AL.MB_CA` and the mushroom body are essentially entirely excitatory, while the optic lobe and the lateral horn are the most inhibitory structures in the brain. This result depends only on counting neurons by their predicted neurotransmitter within each group, and not on any comparison between classes of motif, which is why it survives the difficulties described below.

Figure 17 shows the related and also robust observation that feed-forward edges are more excitatory than feedback ones. That difference, however, is almost entirely attributable to the medulla, whose edges dominate the feed-forward class, and it is a statement about edges rather than about waists.

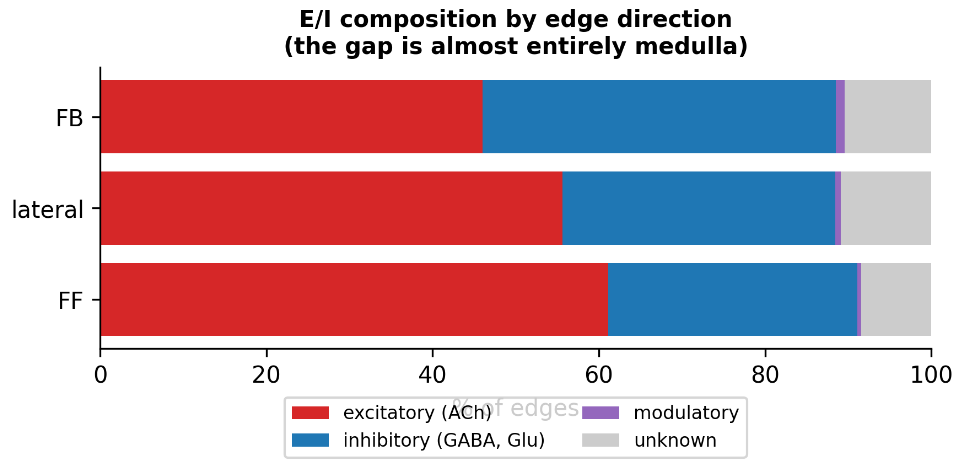

Figure 17: Excitatory and inhibitory composition of edges by direction. Feed-forward edges are more excitatory than feedback ones, but the difference is almost entirely attributable to the medulla. This figure concerns edges; the waist-level comparison that follows, which is the hypothesis actually under test, does not survive replication.

4.7.2 What Does Not

The hypothesis under test was that the waists of feedback bow-ties differ neurochemically from those of feed-forward bow-ties. The natural way to test it is to compare, for each waist group that hosts motifs of both kinds, the excitatory fraction of that group as it appears in the two classes; working at parity of waist in this way controls for the regional composition that Section 1.2 identified as the principal confound.

On the shallow window the effect appears clear, with a gap of $+0.117$ at $p = 0.0033$. Extending the same comparison to all four windows produces Table 9 and Figure 18.

Table 9: Excitatory fraction of the waist, *pure_FB* against *pure_FF*, compared at parity of waist across all four level windows.

Window	Shared waists	*pure_FB*	*pure_FF*	Gap at parity	p
$\{1, 2, 3\}$	86	0.393	0.511	$+0.117$	0.0033
$\{3, 4, 5\}$	221	0.501	0.547	$+0.046$	0.0098
$\{2, 3, 4\}$	185	**0.506**	**0.449**	$\mathbf{-0.057}$	0.0296
$\{1, 2, 3, 4, 5\}$	**492**	0.483	0.504	$+0.021$	**0.098**

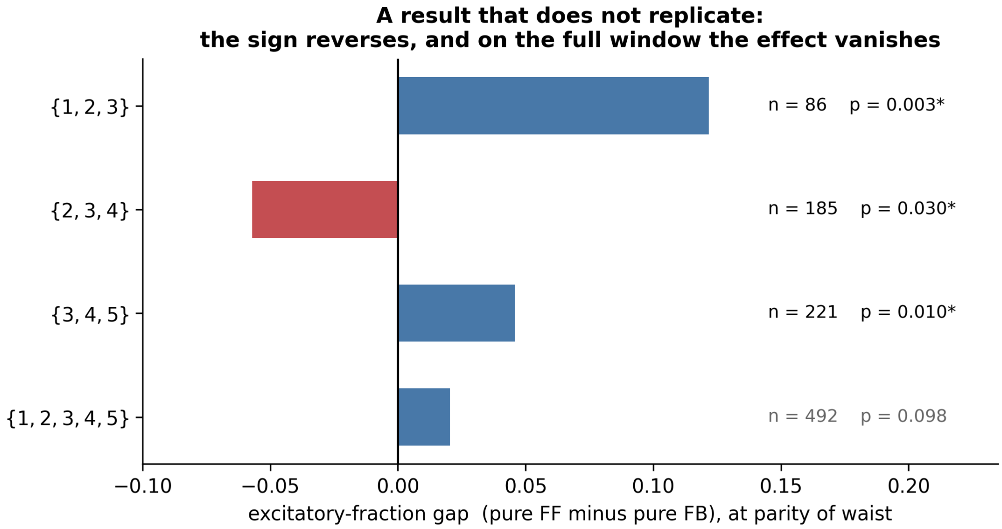

Figure 18: Excitatory fraction gap between *pure_FF* and *pure_FB* waists, measured at parity of waist, across all four level windows, with asterisks marking $p < 0.05$. Window $\{2, 3, 4\}$ falls on the opposite side of zero from the others, and the full window, which has the most shared waists and subsumes the rest, does not reach significance.

Three observations together dismantle the conclusion. On window $\{2, 3, 4\}$ the sign of the effect reverses, with a p-value that would conventionally be called significant: there, feedback waists are *more* excitatory rather than less. On the full window, which has the largest number of shared waists and which subsumes all the others, the effect does not reach significance at all. And the raw gaps on those two larger windows, before any control is applied, are essentially zero, at $+0.0003$ and $+0.0021$ respectively, against $+0.726$ on the shallow window; there is consequently not even an effect present to be explained.

4.7.3 Why the First Window Looked Convincing

The explanation lies in where the two classes of motif happen to sit. On the shallow window the *pure_FF* motifs concentrate in `AL.MB_CA`, which is essentially fully excitatory, while the *pure_FB* motifs concentrate in the optic lobe, which is inhibitory. The apparent difference between motif classes is thus a difference between brain regions wearing the costume of a difference between circuit types.

The control at parity of waist did substantial work: it removed 84% of the raw gap on that window, taking it from $+0.726$ down to $+0.117$. What it did not do was remove all of it, and the residue looked like a genuine effect. In hindsight the residue was the remaining part of the same regional artefact, not an independent phenomenon, and the way this became visible was replication rather than any refinement of the control itself.

It is also worth noting the multiple-testing situation. Across four windows and six pairwise comparisons there are twenty-four tests in total, so roughly one false positive is expected at a nominal $\alpha = 0.05$. None of the four p-values in Table 9 survives a Bonferroni correction within its own window, where the threshold would be 0.0083, except the one obtained on the shallow window.

4.7.4 How This Should Be Read

This is an analysis with a negative outcome: the hypothesis does not hold on the full connectome. It remains a contribution for two reasons. It excludes a plausible hypothesis that would otherwise remain open, and the methodology that exposed the false positive is reusable. That methodology consists of controlling at parity of waist, decomposing the raw gap to see how much of it the control removes, and then replicating across independent subsets of the data. The last of these was decisive: two concordant windows are not a replication but two draws, and it was only the third and fourth windows that settled the question.

4.8 Integrative, Broadcast and Mixed Networks

Allowing k_{in} and k_{out} to vary independently distinguishes three regimes. A circuit with many inputs and one output is *integrative*, gathering information from several sources into one channel; a circuit with one input and many outputs is a *broadcast*, distributing one signal widely; and balanced configurations are mixed.

Enumeration is again avoided, this time by a different route. The sum over all k-combinations of a product of flow vectors is the elementary symmetric polynomial e_k of those vectors, which can be evaluated in $O(p \cdot k)$ time per neuron for p available groups without ever forming a single combination explicitly. The complete sweep for k from two to five on the shallow window takes one hundred and eleven seconds, where explicit enumeration would have required forming 2.9 million combinations for a single waist at $k_{\text{in}} = 3$.

One methodological caution applies to the resulting index, and it is instructive because the artefact is easy to miss. The raw integration index is confounded with the number of groups available to a waist, since e_k over p groups grows combinatorially with p: its Spearman correlation with $\log(p/q)$, where q is the number of output groups, is $+0.535$ at $p = 3.6 \times 10^{-38}$. A waist with many neighbours would therefore appear integrative purely on that account. Normalising by the number of available combinations reduces the correlation to -0.023 at $p = 0.61$, which eliminates the artefact, and only the normalised index is interpreted here.

4.8.1 The Sweep Recovers the Mushroom Body Architecture

Across the four hundred and ninety-seven metanodes above threshold the classification yields one hundred and eighty-four integrative circuits, two hundred and fifteen mixed and ninety-eight broadcast. What makes this more than a tally is that the assignments trace a known anatomical pathway without having been given any information about it.

Table 10: Normalised integration index along the olfactory pathway. Negative values indicate broadcast, positive values integration.

Waist	in / out	Index	Regime
`AL.MB_CA_L2` (projection neurons)	46 / 99	−0.895	broadcast
`AL.LH_L2`	100 / 192	−0.712	broadcast
`MB_CA.MB_ML_L3` (Kenyon cells)	58 / 53	+0.061	mixed
`MB_ML.CRE_L2` (MB output)	45 / 47	+0.997	integrative
`MB_PED.CRE_L2`	28 / 29	+1.000	integrative

The sequence in that table is the textbook description of the mushroom body. Olfactory projection neurons broadcast onto a much larger population of Kenyon cells, which expands a compact odour representation into a sparse high-dimensional one; the Kenyon cells themselves operate in a mixed regime; and the mushroom body output neurons integrate that expanded representation back down into a small number of channels that drive behaviour. The analysis is purely structural and was not seeded with any anatomical knowledge, so recovering this organisation is best understood as a validation of the index rather than as a new biological claim.

4.9 Within-Area Characterisation

The analyses so far have studied interactions *between* anatomical areas. A complementary question is what the motif composition looks like *within* a single area, and whether areas differ from one another in that respect.

An area here is a neuropil, and a neuron is taken to belong to every neuropil named in its group assignment, so that a neuron labelled `AL.MB_CA` counts towards both the antennal lobe and the mushroom body calyx. Areas therefore overlap, which is anatomically correct: a projection neuron genuinely does arborise in both structures. Of the forty-four neuropils in the dataset, thirty-five contain at least two hundred neurons and are analysed here.

Counting within a single area raises a difficulty that does not arise between areas. Reciprocal connections within one neuropil are frequent, so the sets of neurons available as inputs and as outputs overlap heavily, and the requirement that the five participating neurons be distinct can no longer be handled by simply choosing disjoint groups. The count is therefore computed in closed form with an inclusion–exclusion correction that subtracts the configurations in which neurons coincide. Explicit enumeration would be quite infeasible, requiring on the order of 10^{10} combinations in the medulla alone, whereas the closed form processes the entire connectome in 4.5 seconds. The formula was verified against exhaustive enumeration on three hundred random graphs, with exact agreement on all sixteen signatures.

4.9.1 What Survives the Null, and What the Null Removes

Across five hundred and sixty tests, being thirty-five areas by sixteen signatures, one hundred and one are significant at $q < 0.05$ after Benjamini–Hochberg correction. The comparison worth reporting, however, is not that count but what changes between the naive analysis and the controlled one.

Table 11: Enrichment before and after applying the within-area null model. The naive comparison against the global profile produces large values that the null removes entirely.

Area	Naive enrichment	Versus null	q
`EB`	12.6×	**1.00**×	1.00
`MB_CA`	—	0.76×	1.00
`NO`	13.5×	1.99×	0.028
`FB`	7.3×	1.84×	0.028

The ellipsoid body falls from an apparent twelve-fold enrichment to exactly one-fold: the entire effect was a consequence of how that area's neurons are distributed across levels, and the null neutralises it completely. The noduli and the fan-shaped body retain a smaller but genuine enrichment of roughly twice the null expectation. The null model therefore does both of the jobs asked of it, removing false positives while preserving real signals, which is the property that makes the surviving results worth reporting.

Three findings survive the control. The antennal lobe shows 31% of its motifs in the `lat->lat` class, with its neurons spread across several levels rather than concentrated in one, and it is the only genuinely informative lateral case in the dataset. It corresponds to the network of local interneurons that performs lateral inhibition between olfactory glomeruli, which is a well-established circuit recovered here independently and not explicable by level concentration. The optic lobe shows the re-entrant medulla pattern at 57.3% and the lobula plate relay at 40.9%, consistent with Section 4.4.1 and reached by an entirely independent route. The associative areas, by contrast, show no comparable structure, which is itself informative: the lateral organisation observed elsewhere is specific to particular circuits rather than a generic property of dense tissue.

4.10 Computational Performance

Table 12 reports the effect of the sparse reformulation on the two most expensive waists of the search.

Table 12: Effect of the sparse reformulation on the most expensive bottlenecks.

Bottleneck	Before	After	Speed-up
`ME_L2` ($\lvert B\rvert = 24{,}835$)	3172 s	14.5 s	219×
`ME_L3` ($\lvert B\rvert = 10{,}709$)	355 s	1.4 s	254×

The single waist `ME_L2` previously accounted for roughly half of the total runtime on the shallow window, which is what makes a change of this magnitude on one bottleneck worth the effort. With the sparse engine in place, no waist falls back to the reference loop on any of the deeper windows. End to end, the shallow window went from one hour and forty-four minutes to fourteen minutes and thirty-nine seconds.

Output distillation, described in Section 3.5.3, reduced the four windows from over 100 GB to 68 MB, a reduction of up to 18,847-fold on the largest window, while producing downstream results that are identical to those obtained from the full output. The distillation itself costs approximately thirty per cent of the search time, since the aggregates must be maintained as the rows are produced; this is the price of not writing the intermediate at all, and it is worth paying only because the search has become fast enough that regenerating the rows costs less than storing them.

5 Conclusions

5.1 Summary of Contributions

This thesis analysed the bow-tie architecture of the adult *Drosophila* connectome at two complementary scales, and its contributions fall into four groups.

The theoretical contribution is the characterisation of when motif counts compose, stated as Proposition 3.1. The boundary between linear counting and the $\Sigma\Pi$ formulation is the presence of branching, and cyclic patterns require quadratic memory and are therefore out of reach at connectome scale. The result is verifiable by exhaustive enumeration, was so verified on all thirty-one patterns up to four nodes, and applies beyond this particular connectome to any counting problem of the same shape. The accompanying catalogue also places the method honestly: already at four nodes, 87% of connected patterns fall outside what it can handle.

The computational contribution is to have made exhaustive counting tractable. Reformulating the inner loop as a matrix product delegated to optimised libraries, and then switching to a sparse product on the large waists where the dense form fails, produced a factor of 219 on the worst bottleneck and reduced the shallow window from one hour and forty-four minutes to under fifteen. Recognising that the row-level output is an intermediate rather than a result reduced storage from over 100 GB to 68 MB with no loss to any analysis.

The empirical contributions concern the architecture itself. The hourglass property is present, with a score of 0.498 under structural routing, but it depends on the routing scheme and is markedly weaker than in *C. elegans*, plausibly because the fly maintains several parallel modality-specific bottlenecks rather than one shared one. The deep brain is laterally dominated, with purely lateral branches carrying 76.2% of the motif mass against 20.1% in the shallow brain. The deepest level of the hierarchy is the most internally dense and yet marginal as a waist, a combination that points to recurrent rather than convergent organisation. And the two scales of analysis agree on the mid-hierarchy olfactory waists while diverging interpretably on input groups.

The methodological contribution is the negative result together with the procedure that established it. An apparent neurochemical difference between feedback and feed-forward waists did not survive replication across independent level windows and was traced to regional composition. The related finding that the ranking between signatures inverts between null models makes the same point from a different direction: single-null analyses of this kind can be confidently wrong.

5.2 Limitations

The hierarchical decomposition on which everything else rests is inherited from earlier work rather than derived here. It is validated independently in Section 4.1, where its functional composition follows the expected sensory-to-motor gradient, but any result conditioned on levels inherits whatever assumptions went into producing them.

The hourglass score depends on the routing scheme, taking values between 0.247 and 0.498 depending on the scheme and the path-length bound. No single number should be quoted in isolation, and the comparison with *C. elegans* should be read with the same caution, since the two studies do not use identical routing conventions.

The coarse motif classification is close to vacuous, with 94.4% of patterns falling into the *mixed* bucket. The sixteen-signature decomposition addresses this, but even after it the residual `mix->mix` class retains 29.5% of patterns in the shallow window. Resolving that class further would require distinguishing branches by which specific levels they connect, at the cost of a considerably larger label space.

No null model isolates the hypothesis cleanly. The three used here preserve different constraint sets and none is fully satisfactory: model A preserves global degrees but destroys anatomy, model B preserves anatomy but destroys degrees, and model C preserves both while having a ratio dominated by sparsity. A null that simultaneously preserved within-block degrees and block densities would make the statistic essentially deterministic, since the motif count is very nearly a deterministic function of the degree sequences. The limitation is intrinsic to the quantity being tested rather than to the implementation, and Section 4.5.1 shows that it has real consequences for interpretation.

Realised compression remains sensitive to a collapsing waist. When the flow is carried by a single neuron the ratio explodes, with `LO_L3` reaching 6316 at an effective waist of 1.3 and `ME_L2` reaching 4259 at 1.8. Such structures are hubs rather than waists and are excluded here by a threshold on the effective waist size, set to five throughout, but that threshold is a declared choice and results should be checked for robustness against it. Reporting the pair consisting of the effective size and the periphery, instead of their ratio, would avoid the division entirely and is worth considering.

The pattern catalogue stops at four nodes, since exhaustive enumeration on five would require 2^{20} edge masks over 120 permutations. The classification of Proposition 3.1 does not depend on pattern size, so the qualitative conclusion carries, but the proportion of non-factorisable patterns beyond four nodes is not measured here. Relatedly, that proposition concerns homomorphisms, whereas the requirement that participating neurons be distinct is a separate inclusion–exclusion layer; the two should be kept conceptually apart.

The sweep over k_{in} and k_{out} covers only the pure forms. Shapes with both parameters at three or more would require inclusion–exclusion to a depth this implementation does not reach. The trichotomy of Section 4.8 does not need them, since integrative and broadcast circuits are precisely the pure forms, but a complete map of the parameter plane would.

The distillation fixes the selection criteria at search time, since the retained top-N selections are chosen by occurrence count. An analysis requiring a different criterion, such as selection by realised compression, would require re-running the search rather than merely re-reading the stored output.

Finally, and most fundamentally, all the results here are structural. Connectivity constrains computation but does not determine it, and no claim about neural dynamics follows directly from these counts. What a structural analysis can establish is which circuits exist and in what abundance; what they do remains a question for physiology.

5.3 Future Directions

The most immediate extension is to carry the remaining analyses through to the deep windows. Realised compression and statistical validation were conducted in depth only on the shallow window, which is the least rich of the four with thirty-one million motifs against four hundred and thirty-four million, and there is no methodological obstacle to repeating them now that the search is fast enough.

The behaviour of level 5 deserves separate investigation. It is the most internally dense level of the connectome and yet marginal as a waist, and characterising the connectivity that produces this combination, presumably recurrent and lateral rather than convergent, is a well-posed structural question with a plausible functional interpretation attached to it.

The framework already supports arbitrary values of k_{in} and k_{out}, so a systematic exploration of asymmetric configurations would connect the bow-tie analysis to the integrative and broadcast distinction of Section 4.8 in a more thorough way than the present sweep allows.

Finally, both the counting framework and the hourglass measurement are species-agnostic. Applying them to other fully mapped connectomes as those become available would establish whether the architectural principles found here are general or particular to the fly, which is the question that motivates structural connectomics in the first place.

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